

Fifty Bacterial Datasets for *E. coli* and *P. putida*, Ranked by Scale and Density

Michael Volk

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The database holds 51 schematized and L0–L4-verified *Saccharomyces cerevisiae* datasets and no bacterial dataset at all. Table 5 ranks 75 candidates for the two bacterial hosts, *Escherichia coli* and *Pseudomonas putida*, by measurements, meaning instances times phenotype dimensionality (Sec. 1); the first fifty are the recommended builds and the remaining 25 are a ranked reserve. The build is staged in two tranches, twenty rows then thirty, with a floor of six rows per host in the first tranche so that neither host’s setup cost waits on the other (Sec. 1.5). Of the fifty, 39 are *E. coli* and 11 are *P. putida*.

One consequence of that ordering is large enough to state before the table rather than after it (Sec. 1.2). The rows that connect an engineering intervention to a production phenotype carry a median of 145 measurements against 282,906 for the functional-genomics screens, so twenty-one of the twenty-four engineering rows fall into the reserve. The ranking is reported as specified, and the records carry the class field a portfolio rule would need to reorder them.

Two changes to the schema gate every row, and both were established by running the validators rather than by reading them (Sec. 3): the gene-identifier namespace rejects every bacterial gene name, and the genomes tier has no bacterial assembly set. A third thing expected to block turns out not to, since the reference genome record already accepts either host unchanged.

Table 7 is the column that answers whether these rows fit the record types already in use: for each one, the built yeast dataset it mirrors and what the loader would still need. Table 6 gives each row’s citation, link and data location; Table 8 records what was considered and dropped; Sec. 8 closes with the summary statistics and the confidence split.

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1 The selection rule ✗

1.1 Rows are ordered by measurements, which is scale times density ✗

The order is one quantity, applied to every row alike: **measurements**, meaning instances times phenotype dimensionality, descending. **Instances** is the number of genotype $\times$ environment $\times$ replicate records a row can actually yield, and **dimensionality** is how many values each of those records carries. A scalar fitness screen of n strains contributes n numbers; a proteome panel of n samples at a depth of 2,000 proteins contributes $2 \times 10^3 n$.

Ranking on instances alone gets the order visibly wrong, and it is worth naming the failure rather than asserting the rule. A quantitative proteome across a couple of dozen growth conditions holds fewer instances than a small deletion screen, which would rank it below one, and yet each of its instances is a complete proteome. The product is what places it correctly, and it is the same normalization the supported-datasets table already applies through its compressed-signal column and the same one the yeast candidate list ranks inside its bands with.

Two tie-breaks make the rule total. Instances break a measurement tie, so between two rows holding the same number of values the one spanning more distinct strain and condition combinations wins; a wide, shallow row is worth more to a genotype-to-phenotype model than a narrow, deep one of equal volume. Tier breaks that.

This differs from the yeast list on purpose. There, rows are banded first by what they supply to a planned single-cell campaign, and scale orders only inside a band. Here scale and density are the requested rule directly, so no band is applied and no row is promoted for what it pairs with. What a row combines with is still recorded, in Table 7 and in the join column, but it does not move the row.

1.2 What this ordering costs, stated before the table ✗

Ranking on measurements has one large consequence and it runs against the reason a bacterial host is interesting at all. The functional-genomics screens win, and the metabolic-engineering campaigns lose, by roughly three orders of magnitude.

The twenty-four rows that connect an explicit engineering intervention to a production phenotype, meaning the production campaigns, the combinatorial designs and the tolerance series, have a median of 145 measurements and a median rank of 62. Three of them reach the fifty; twenty-one sit in the reserve. The sixteen transposon-fitness and chemical-genomics rows have a median of 282,906 measurements, a factor of about 1,944 more, and every one of them reaches the fifty. Twenty-one of the twenty-five reserve rows are engineering rows.

Neither figure is wrong and the rule is not misapplied. A titer measured on six strains really is six numbers, and a barcoded library really does yield millions. What the ordering encodes is that a dataset's worth is its volume, and for a substrate meant to support inverse strain design that is a defensible default and an incomplete one: the rows that teach a model what an intervention does to a product are the rows this ordering ranks last.

Two things follow, and both are now done. The ranking is reported as asked, so the measurement order is recoverable for every row and is printed beside the final order in Table 1. And the correction turned out to be one field rather than a rewrite: a per-row product axis, pinned after ranking, reorders the same records without recollecting anything. Section 2 applies it to the one product the work targets, where the three-order-of-magnitude penalty had put four of eight rows in the reserve. The same mechanism extends to any other product that becomes a priority, which is the argument for a pin over a reweighted score.

1.3 Sequence reconstruction is the gate, not a preference ✗

A phenotype is only trainable against a genotype that can be written down, so every row names a **sequence basis**: the route by which the total genomic content of one strain would be reconstructed. The bases are named per host, because the reference differs and because a b-number and a PP_ locus tag are different namespaces that must not be silently merged (Sec. 3).

- **K-12-KO** and **KT2440-KO** – the reference genome minus one cataloged gene. The cheapest basis, and the one most of the built yeast set sits on in its own reference.
- **K-12-KO x KO** – a constructed double mutant, both loci cataloged, which is the basis a genetic-interaction row needs.
- **K-12+transposon** and **KT2440+transposon** – a mapped insertion site carrying a barcode. The insertion position is sequenced, so the genotype is realized rather than designed, which is what separates a barcoded transposon library from a guide library.

- **K-12+guide**, **KT2440+guide** – the genome is unedited and the perturbation is a designed cassette plus a guide target. Regulatory, not sequence-level, and it reaches the essential genes a deletion collection cannot hold.
- **KT2440+promoter** and **engineered-chassis+promoter** or **+RBS** – a designed regulatory part over a native or heterologous gene. Designed rather than verified per strain.
- **engineered-chassis** – a named production strain plus heterologous cassettes, which is what a titer is measured on.
- **evolved-WGS** – a resequenced evolved clone.
- **reference-only** – wild type, where the environment carries the perturbation.

The *E. coli* bases say **K-12 rather than naming one strain, and that is an admission**. The deletion collection is in BW25113 and the knockdown libraries are mostly MG1655. Those are different genomes: BW25113 carries defined lesions at the lactose, arabinose and rhamnose loci among others, so a deletion row and a knockdown row do not sit on the same reference even when both are called K-12. Labeling a collection row with the MG1655 name would be a sequence-level error in a substrate whose claim is sequence-level genotype fidelity, and labeling both K-12 records the lineage without asserting an identity that was not checked. Which background each row actually used is a per-row provenance item to settle when its loader is written, and it is not settled here.

What fails the gate is a strain whose genome is unknown even in principle. An evolved population that was never resequenced lands there, and so does a strain described only by a phenotype in a review table. Those are dropped in Table 8 rather than ranked low, because no amount of phenotypic value substitutes for a missing genotype.

1.4 Tier, and what it is for here ✖

Tier no longer orders the list, so it does one job: it records how much of a dataset a loader would actually get, which is what makes a high measurement count trustworthy or not.

- **Tier 1** clears all three of $\geq 10^3$ genotypes, $\geq 10^4$ instances, and a clean sequence basis.
- **Tier 2** clears two of the three, or is the only dataset covering a phenotype the substrate needs for these hosts.
- **Tier 3** is a modality or reference backbone the substrate needs without itself carrying a large genotype axis, such as a library definition or a per-gene coefficient that applies to every strain alike.
- **Tier 4** is real but small, coarsely labeled, or expensive to extract.

The bars apply to what a row contains rather than to its reputation. A curated corpus with thousands of literature records fails tier 1 whatever its size, because its records re-serve primary papers and the net-new fraction is unknown until it is split by source.

1.5 Staging, and why the first tranche carries a floor ✖

The fifty are built in two tranches, twenty then thirty. The second tranche is the next thirty by the ordering rule and needs no further explanation. The first twenty carry two corrections to the rule, the isoprenol pin of Sec. 2 and the per-host floor below. Both are applied after ranking and reported rather than folded into a score.

The correction is a floor of six rows per host, not an even split, and the difference matters. The two schema changes in Sec. 3 are per host, and so is provisioning a reference release, so neither cost is paid until a row for that host is actually built; a floor guarantees both hosts start. An even ten-and-ten split would go much further than that and fight the ordering rule, promoting four rows of a few hundred measurements over rows of a few hundred thousand.

On these rows the floor does not bind. Ranking by measurements alone already puts fourteen *E. coli* rows and six *P. putida* rows in the first tranche, which meets the floor without promoting anything, and the isoprenol pin then raises the *P. putida* count further because that host carries most of the isoprenol work. Every promotion the floor makes is printed, so a future pass in which it does bind will say so rather than quietly reordering. An even ten-and-ten split would have bound here, which is the comparison that justifies the floor over a split.

2 Isoprenol, and the pin that the ordering rule needs ✖

2.1 The product named as the priority is the one the rule ranks last ✖

Isoprenol is 3-methyl-3-buten-1-ol, an aviation fuel precursor, and it is the product the bacterial strain-design work targets. It appears in the literature under two names: the earlier papers write **isopentenol** and the current ones write

isoprenol, and they are the same molecule, so a row counts as measuring it under either word.

Section 1.2 stated the cost of ordering rows by measurements in general terms. Isoprenol is the case where that cost becomes a concrete defect. A titer campaign measures tens to a few thousand numbers, because a strain yields one titer per condition per replicate and strains are built by hand. A barcoded fitness screen measures 10^6 to 10^8 . No isoprenol campaign can outrank any screen, whatever its relevance, so measurement order alone placed four of the eight isoprenol rows outside the fifty recommended builds, at ranks 57, 64, 69 and 70 of 75. The highest-priority product in the portfolio sorted into the reserve.

2.2 A pin, reported as moves, rather than a tuned weight ✗

Two repairs were available. A weight on production relevance folded into the score would reorder the table, and would also make the ordering unauditable: a reader could no longer recover what scale alone said, and the weight would be chosen by whoever wanted a particular answer. A pin applied after ranking does the same reordering and stays reversible, so that is what the generating script does.

The axis divides by the role the molecule plays in the experiment, because that is what decides whether a row teaches production or teaches survival.

- **direct** – isoprenol or isopentenol is the measured product, meaning a titer. Lifted into the first tranche.
- **tolerance** – isoprenol is the stressor and the readout is growth, with no titer. Lifted into the first tranche as well, since the molecule of interest is still the one being measured against. A row reporting both a titer and a tolerance assay is direct, because a row carries one value and the titer is the stronger label.
- **precursor** – the measured product is a different isoprenoid or terpenoid, so the row carries the mevalonate or MEP pathway that feeds isoprenol without carrying the product. Lycopene, pinene, mevalonate and geranic acid are here. Lifted above the cut at fifty, not into the first tranche.
- **analog** – the stressor is a short-chain alcohol or a biomass inhibitor standing in for isoprenol rather than isoprenol itself. Reported on the axis and *not* lifted. Substituting one alcohol for another assumes cross-tolerance, and a row is not promoted on an assumption.

Each lift displaces the weakest row inside the boundary that no pin protects, and every lift is recorded with its rank before and after. Table 1 prints the rank the measurement rule gives on its own beside the rank this document prints, so the pin can be undone row by row. The per-host floor of Sec. 1.5 reads the same protection, which is what keeps one mechanism from canceling another.

2.3 What the sweep found, and what it missed ✗

A metadata sweep over Europe PMC returned 300 publications for the two hosts, 200 *E. coli* and 100 *P. putida*, and it is held as a separate artifact with a separate generating script because none of its record counts have been verified. Every cell of its instance column reads “Verify from supplement”, and its own overview states the limit: “this is a 300-candidate discovery queue, not 300 download-verified datasets.” Table 2 gives its composition and Table 3 every isoprenol-relevant row it returned. A queue row is a lead, and promoting one means fetching the paper, counting its axes and writing a curated row; nothing in the queue may be cited as though its scale were known.

The sweep flagged six publications as measuring isoprenol directly, four for *P. putida* and two for *E. coli*. All four *P. putida* papers were already curated here, which is a useful independent check on coverage. Both *E. coli* papers were absent and are now added, and they matter out of proportion to their size: before them the table held no isoprenol row for *E. coli* at all, so the product existed in one host only and no cross-host transfer could be posed.

The sweep also missed two isoprenol papers this table already held, a CRISPRi downregulation campaign and a lignocellulosic-sugar conversion campaign that reports isoprenol alongside isoprenyl acetate. Both measure isoprenol titer and neither carries the word in its title, which is the mechanism of the miss: title and abstract matching finds a paper named for its product and not one named for its method. Two misses out of eight is a 25 percent false-negative rate on the one axis the sweep was built to find, measured against a list assembled by hand, so the queue is a source of leads and not a measure of what exists.

Table 1: The isoprenol axis, divided by the role the molecule plays. *Rule* is the rank the measurement ordering gives on its own and *Final* the rank this document prints, so the pin can be undone row by row. *Direct* measures isoprenol or isopentenol as a product, the same molecule under two names; *tolerance* measures growth against it as a stressor; both are lifted into tranche 1. *Precursor* measures another isoprenoid and is lifted above the cut. *Analog* substitutes a different alcohol or inhibitor and is reported but not lifted, because the substitution assumes cross-tolerance.

Dataset	Host	Axis	Product	Rule	Final	Lifted by
Carruthers 2025	<i>P. putida</i>	direct	isoprenol	6	6	–
Yunus 2026	<i>P. putida</i>	direct	isoprenol	72	14	isoprenol
de Siqueira 2025	<i>P. putida</i>	direct	isoprenol	71	15	isoprenol
Lim 2025 isoprenol TALE	<i>P. putida</i>	direct	isoprenol	65	17	isoprenol
Tian 2019 isopentenol CRISPRi	<i>E. coli</i>	direct	isopentenol	63	18	isoprenol
Kang 2026 isoprenyl acetate	<i>P. putida</i>	direct	isoprenyl acetate, isoprenol	57	19	isoprenol
Menasalvas 2025	<i>P. putida</i>	direct	isoprenol	50	20	isoprenol
Wang 2015 isoprenol tolerance	<i>E. coli</i>	tolerance	isoprenol	67	16	isoprenol

Table 2: The discovery queue by class and host. *Curated* counts rows already present in Table 5, matched on DOI, so the overlap is measured rather than asserted. *Direct* and *Adjacent* are the source’s isoprenol axis. No row carries an instance count, because the source asserts none.

Class	<i>E. coli</i>	<i>P. putida</i>	Curated	Direct	Adjacent
Joint	35	8	5	3	4
ME breadth	115	67	7	2	13
Scale anchor	50	25	3	1	0
Total	200	100	15	6	17

Table 3: Every isoprenol-relevant publication the sweep returned, the source’s own flag. *Direct* measures isoprenol or isopentenol itself; *adjacent* measures another isoprenoid or terpenoid, so it carries the precursor pathway without the product. *Curated* names the row in Table 5 when the paper is already there, and is otherwise the remaining work.

Flag	Host	Year	Title	Curated
direct	<i>E. coli</i>	2019	“Redirecting Metabolic Flux via Combinatorial Multiplex CRISPRi-Mediated Repression for Isopentenol Production in Escherichia coli.”	–
direct	<i>E. coli</i>	2015	“Dynamic interplay of multidrug transporters with TolC for isoprenol tolerance in Escherichia coli.”	–
direct	<i>P. putida</i>	2025	“Alternate routes to acetate tolerance lead to varied isoprenol production from mixed carbon sources in Pseudomonas putida.”	de Siqueira 2025
direct	<i>P. putida</i>	2025	“Automation and machine learning drive rapid optimization of isoprenol production in Pseudomonas putida.”	Carruthers 2025
direct	<i>P. putida</i>	2025	“Biosensor-driven strain engineering reveals key cellular processes for maximizing isoprenol production in Pseudomonas putida.”	Menasalvas 2025
direct	<i>P. putida</i>	2025	“Evolution-guided tolerance engineering of Pseudomonas putida KT2440 for production of the aviation fuel precursor isoprenol.”	Lim 2025 isoprenol TALE
adjacent	<i>E. coli</i>	2013	“Chromosomal evolution of Escherichia coli for the efficient production of lycopene.”	–
adjacent	<i>E. coli</i>	2021	“Fine tuning the glycolytic flux ratio of EP-bifido pathway for mevalonate production by enhancing glucose-6-phosphate dehydrogenase (Zwf) and CRISP...”	–
adjacent	<i>E. coli</i>	2022	“Enhancing the Production of Pinene in Escherichia coli by Using a Combination of Shotgun, Product-Tolerance and I-SceI Cleavage Systems.”	–
adjacent	<i>E. coli</i>	2019	“Genomic and transcriptional changes in response to pinene tolerance and overproduction in evolved Escherichia coli.”	–
adjacent	<i>E. coli</i>	2023	“Evaluation of Various Escherichia coli Strains for Enhanced Lycopene Production.”	–
adjacent	<i>E. coli</i>	2024	“Rational engineering of Escherichia coli strain for stable and enhanced biosynthesis of pinene.”	–
adjacent	<i>E. coli</i>	2025	“Increased Mevalonate Production Using Engineered Citrate Synthase and Phosphofructokinase Variants of Escherichia coli.”	–
adjacent	<i>E. coli</i>	2026	“Metabolic engineering of endogenous MEP pathway for enhanced lycopene production in Escherichia coli.”	–
adjacent	<i>E. coli</i>	2025	“Enhanced isoprenoid production in Escherichia coli cells harboring the archaeal mevalonate pathway.”	–
adjacent	<i>E. coli</i>	2025	“Synergistic Production of Lycopene and -Alanine Through Engineered Redox Balancing in Escherichia coli.”	–
adjacent	<i>E. coli</i>	2026	“Heterologous Production of Torularhodin, the Monocyclic Carotenoid with a Terminal Carboxyl Group, in Escherichia coli.”	–
adjacent	<i>E. coli</i>	2026	“Development of a transcription factor-based biosensor strain for reporting -terpineol production via the alcohol-dependent hemiterpene pathway in ...”	–
adjacent	<i>P. putida</i>	2019	“Mevalonate production from ethanol by direct conversion through acetyl-CoA using recombinant Pseudomonas putida, a novel biocatalyst for terpenoid ...”	–
adjacent	<i>P. putida</i>	2014	“De novo production of the monoterpene geranic acid by metabolically engineered Pseudomonas putida.”	–
adjacent	<i>P. putida</i>	2022	“Engineering isoprenoids production in metabolically versatile microbial host Pseudomonas putida.”	–
adjacent	<i>P. putida</i>	2026	“Characterisation of metabolic burden in Pseudomonas putida reveals precursor limitation in heterologous lycopene production.”	–
adjacent	<i>P. putida</i>	2019	“Engineering Pseudomonas putida for isoprenoid production by manipulating endogenous and shunt pathways supplying precursors.”	–

3 What the schema needs first ✗

Two changes gate every row in Table 5, and a third thing that was expected to gate them does not. All three were established by running the validators rather than by reading them, because the question is what the code accepts, not what it appears to accept.

Table 4: What the schema needs before any bacterial row can be written, and how each was established. The first two are the critical path: every row in Table 5 waits on them, and both are additive, so they add classes and move no served closure. *Evidence* is the result of running the validator, not a reading of it.

Change	Where	Evidence
A bacterial gene-identifier namespace on GenePerturbation, so a b-number or a PP_ locus tag validates as a systematic gene name	<code>torchcell/datamodels/schema.py</code> , <code>GenePerturbation.validate_sys_gene_name</code>	The validator admits only Y[A-P][LR]NNN[WC], QNNNN and YNC[A-Q]NNNN[WC]. Constructing a DeletionPerturbation with b0002, PP_0001, ECK0002 or thrA raises 'Invalid systematic gene name format'; YAL001C is accepted. Blocks: every row in the table that perturbs a named gene.
A genomes-tier assembly set per host, so a strain's sequence resolves the way a yeast strain's does	<code>torchcell/sequence/genome/registry.py</code> and <code>torchcell/sequence/genome/</code>	The tier defines two assembly sets, both <i>S. cerevisiae</i> (<code>sgd_S288C_R64-4-1_20230830</code> and <code>peter2018_1011_assemblies</code>), and the genome package holds only a <i>scerevisiae</i> subpackage. Blocks: the sequence basis of every row, and so the sequence-reconstruction gate.
Nothing on the reference side	<code>torchcell/datamodels/schema.py</code> , <code>ReferenceGenome</code>	species and strain are free strings, so <code>ReferenceGenome(species='Escherichia coli', strain='MG1655')</code> and the <i>KT2440</i> equivalent both validate unchanged. Blocks: nothing.

The reference side already admits both hosts. `ReferenceGenome` carries `species` and `strain` as free strings, so `ReferenceGenome(species="Escherichia coli", strain="MG1655")` and the *Pseudomonas putida* *KT2440* equivalent both validate with no change at all. The organism was never hard-coded at the reference; it was hard-coded one level down.

The genotype side rejects every bacterial gene name. `GenePerturbation` validates `systematic_gene_name` against three patterns, and all three are *S. cerevisiae*: the nuclear open reading frames, the mitochondrial genes, and the noncoding genes. Constructing a deletion with the *thrA* b-number, a *P. putida* locus tag, an *EcoCyc* identifier or a bare gene symbol raises *Invalid systematic gene name format* in each case, while a yeast name is accepted. This is not a naming inconvenience. `Genotype` derives its content identity from the sorted systematic names of its perturbations, so the identifier namespace is load-bearing for node identity in the graph, and a b-number and a PP_ locus tag must stay distinguishable namespaces rather than being merged into one string field.

The fix is additive in the sense the admission gate uses: a bacterial namespace accepted alongside the yeast patterns adds a validated form and changes no record any served dataset already holds, so no served closure moves. That is the same shape as the segregant genotype added for the sixteen-cross yeast panel, where a genotype that carried no gene edit at all needed a sibling class rather than a widened one.

The genomes tier has no bacterial assembly set. The tier that dereferences a strain to sequence defines two assembly sets, both yeast, and the genome package holds one organism subpackage. A host needs its reference release pinned there by sha256 before any row's sequence basis resolves to bytes, so this is ordinary provisioning work rather than a design question, and it is on the critical path for the same reason the identifier namespace is.

3.1 What this buys, and what it does not ✗

The two changes above are enough to write a deletion, a transposon insertion, a guide knockdown and a production titer for either host. They are not enough for three things this table contains, and each carries its own cost in Table 7.

A designed combinatorial construct, meaning a ribosome binding site or promoter library over a heterologous pathway, states what each member was built to be rather than what it turned out to be. Ingesting one commits to storing the design and a verification status, which is the same weaker claim the yeast list records for its designed-edit rows.

A resequenced evolved clone is a genotype only where the resequencing exists. An evolved population that was never sequenced differs from its parent at unmapped positions, so it fails the sequence-reconstruction gate and is dropped in Table 8 rather than ranked low.

A curated literature corpus is not one dataset. Its records re-serve primary papers, so the count is not net new until the corpus is split by source identifier, and rows in that state are marked in Table 5 rather than counted as reachable.

4 The fifty, in rank order x

Table 5: The bacterial candidates, ranked by *Meas.* descending, which is instances times phenotype dimensionality (Sec. 1). *Genotypes* and *Env* are the perturbation and condition axes. *Inst.* is genotype×environment records, † where it is the product of the two axes rather than a reported count and ‡ where it is an order-of-magnitude estimate. *Sequence basis* is the route to each strain’s total genomic content; a row with no route is excluded (Table 8). Superscript **B** marks a row whose per-record values are not released, **A** a corpus that re-serves other papers and is not net new until split by source. Citations and data locations are in Table 6; the yeast dataset each row mirrors is in Table 7.

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
1	Fuhrer 2017 Metabolome / flux; non-targeted flow-injection TOF-MS metabolite ion intensities	<i>E. coli</i>	1	3,807 deletions	1 condition	3.4×10^4	2.6×10^8	K-12-KO	3,807 Keio single-gene deletion mutants (from 4,320 after excluding 21 very sick strains and 16 injection failures), two independent clones each in technical duplicate, more than 34,000 mass-spectrometric injections. 7,534 distinct m/z features (3,169 negative plus 4,365 positive mode); 3,130 of them were putatively matched to 1,432 of 2,028 chemical formulas, so annotated depth is far below raw feature depth. Deposited in MassIVE and BioStudies, not MetaboLights.
2	Wetmore 2015 Transposon fitness; barcode competitive fitness (BarSeq)	<i>E. coli</i>	1	152,018 insertion mutants	89 conditions	1.4×10^7 †	1.4×10^7	K-12+ transposon	Table 1 gives 152018 strains with unique bar codes for the <i>E. coli</i> library KEIO_ML9 and fitness estimates for 3471 genes (84 percent). The <i>E. coli</i> lane included 96 samples with 4 time-zero samples across 47 different carbon or nitrogen sources, of which 89 passed quality metrics. Aggregated to gene level the usable record count is 3471 by 89, about 308919. Across all five bacteria the paper reports 387 successful genome-wide assays. No SRA accession is stated in the text; the LBL supplemental URL was taken from the paper and was not fetched.
3	Mutalik 2020 Transposon fitness; barcode competitive fitness (RB-TnSeq BarSeq)	<i>E. coli</i>	1	152,018 insertion mutants	77 conditions	1.2×10^7 †	1.2×10^7	K-12+ transposon	The K-12 BW25113 RB-TnSeq library carries 152018 barcoded insertions in 3728 genes and was assayed against 14 diverse dsDNA phages in 68 RB-TnSeq assays plus 9 no-phage controls, giving the 77 conditions used here. A parallel BL21 library of about 97000 mutants was assayed against 12 phages in 53 pooled fitness experiments plus 9 controls. The same study also ran CRISPRi and Dub-seq arms on the same phage panel, which makes it a three-modality join on one condition axis. Accessions are quoted from the article’s data availability statement; the SRA and figshare pages themselves were not fetched. The related Mutalik 2019 Nature Communications Dub-seq paper (doi 10.1038/s41467-018-08177-8, PMID 30659179) is a barcoded overexpression library of 30558 barcode pairs assayed in 155 fitness experiments over 52 chemicals with SRA PRJNA512427 and figshare 10.6084/m9.figshare.6752753.v1; it is omitted as its own row only because plasmid overexpression has no matching modality value here.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
4	Tong 2020 Fitness / chemical genomics; 24 hour solid-agar colony growth curve	<i>E. coli</i>	1	3,796 deletions	30 conditions	1.1×10^5	8.2×10^6	K-12-KO	Kinetic growth measurements for 3796 genes under 30 carbon source conditions in MOPS minimal medium, 1536 colonies per plate on solid agar, giving 113880 individual growth curves and over 8 million data points. Plates were scanned every 20 minutes over 24 hours, so each instance is a time series of roughly 72 points; the 72 figure is derived from the stated interval and duration rather than stated directly. 51 poorly annotated genes showed a low growth phenotype in at least one carbon source. This is the cleanest released Keio by carbon-source matrix and the only row here carrying a real time axis.
5	PRECISE-1K Transcriptome; RNA-seq expression compendium plus 201 iModulon activities	<i>E. coli</i>	3	76 designs	116 conditions	1.0×10^3	4.4×10^6	K-12-KO	1,035 samples over 4,257 genes, 5 strain backgrounds, 76 unique gene knockouts, 45 distinct projects, yielding 201 iModulons; a combined public K-12 set reaches 2,710 samples and 194 iModulons. The paper reports no single unique-condition total, so 116 is the sum of its itemized condition variables (9 base media, 18 carbon sources, 38 supplements, 42 heterologous proteins, 4 temperatures, 5 pH values). No GEO SuperSeries was found; Zenodo is the stated repository.
6	Carruthers 2025 Production campaign; isoprenol titer by GC-FID plus paired DIA shotgun proteome	<i>P. putida</i>	3	472 guides	1 condition	1.4×10^3 †	2.1×10^6	KT2440+ guide	The full text states 472 unique strains (125 single perturbations and 347 combinations) over six DBTL cycles in triplicate, matching the recalled numbers exactly; machine learning selected roughly 400 constructs out of 800,000 possibilities for a fivefold titer gain. ONE study with eight linked deposits: Dryad 10.5061/dryad.gtht76hzh (proteomic metadata plus a 29.7 MB Top3 peptide quantification table, file list enumerated), a Zenodo and GitHub release of per-cycle titer tables (10.5281/zenodo.17178684, JBEI/Isoprenol_CRISPRi), and seven per-cycle PRIDE accessions PXD063733 (DBTL0), PXD063737, PXD063738, PXD063740, PXD063743, PXD063744, PXD063746 (DBTL6). An ingestion that expects one PXD per paper silently drops six sevenths of the proteomics.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
7	Lim 2022 putidaPRECISE321 A Transcriptome; batch-normalized log2 TPM RNA-seq expression, decomposed into 84 iModulons	<i>P. putida</i>	4	17 wild type	118 conditions	321	1.8×10^6	reference-only	All recalled numbers confirmed twice over, from the paper and by recomputation on the released matrix: 321 samples, 118 unique condition groups, 21 projects, a 5,564 gene by 321 sample log_tpm_norm matrix, 84 iModulons explaining 75.7 percent of variance with 1,265 member genes. The live iModulonDB API returns the dataset under organism p_putida, dataset precise321. Replicate structure recomputed as 40 conditions at n=2, 71 at n=3, 7 at n=4, summing to 321 exactly. Gene-level values total 1,786,044. This row is an AGGREGATION of 21 projects: 541 samples were collected and 321 passed QC, of which the paper says 305 were previously published and 16 newly generated. The genotype count of 17 distinct genotype tuples (10 with an annotated perturbation such as relA, finR, crc, fleQ) is an UNDERCOUNT, because engineered muconate producers and ALE clones are encoded in condition strings rather than genotype fields. There is no umbrella GEO accession: the sample table carries 47 distinct SRA or BioProject accessions covering 216 of 321 samples and 9 GEO Series, and 105 samples carry no accession at all, so per-sample raw-data provenance is incomplete by construction.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
8	Borchert 2024 fModules ^A Transposon fitness; gene-level RB-TnSeq relative fitness, log2 barcode ratio against the time-zero baseline	<i>P. putida</i>	1	4,732 insertion mutants	179 conditions	1.6×10^6	1.6×10^6	KT2440+ transposon	Every recalled number checks out, and the 179 versus 183 discrepancy is real rather than an error: Methods say 183 unique growth conditions were collected, the Abstract says 179 were analyzed, and the released workbook's classifier column has exactly 179 distinct values, so 183 is the collection and 179 is the analyzed matrix. Fitness for 4,732 of 5,564 protein-coding genes. The instance count is VERIFIED, not a product: fModule_Metadata.xlsx (23.3 MB) was opened and its fitness sheet counted at 4,732 rows by 332 value columns with zero nulls, alongside a t-statistic sheet of identical shape. This row is an AGGREGATION and a superset of the other transposon rows: 254 of 332 samples were pulled from the Fitness Browser and 78 are the Beckham-group sets (Putida_ML5_set100 and set101) deposited as SRA PRJNA809672 (15 runs), PRJNA856070 (57 runs) and PRJNA1011287 (39 runs), all counted. Replicate structure: 194 samples from duplicate conditions, 78 from triplicate, 60 singletons. The environment axis is two-slot (condition_1 plus condition_2 with units), which is what keeps glucose-plus-stressor tolerance samples distinguishable from sole-carbon catabolism samples; collapsing it to one string destroys that contrast. Caveat: genes lacking a value in any sample were dropped, so condition-specific genes reported by the primary papers are silently absent here.
9	Caglar 2017 Multi-omics campaign; matched RNA-seq and shotgun proteomics plus central carbon fluxes	<i>E. coli</i>	4	1 wild type	34 conditions	322	1.4×10^6	reference-only	34 growth conditions in <i>E. coli</i> B REL606 with 152 RNA-seq samples, 105 proteomics samples and 65 flux measurements (322 total), spanning exponential and stationary phase plus two time courses from 3 hours to 2 weeks. The paper reports 4,196 distinct mRNAs and proteins as a joint figure, so treat 4,196 as the matched transcript-and-protein gene axis rather than as two independent depths.
10	Nichols 2011 Fitness / chemical genomics; colony size drug-gene fitness score	<i>E. coli</i>	1	3,979 deletions	324 conditions	1.3×10^6 †	1.3×10^6	K-12-KO	The paper states 3979 mutant strains passing quality control, 324 conditions covering 114 unique stresses, and a total of 13497 significant phenotypes at 5 percent FDR, which is about 1 percent of all condition-gene pairs tested. The URL printed in the paper, http://ecoliwiki.net/tools/chemgen/ , now returns HTTP 410 Gone; the same portal is live at the ecoliwiki.org host and serves the full per-strain by per-condition flat file. The Keio background is BW25113, not MG1655 itself.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
11	Goodall 2018 Transposon fitness; TraDIS insertion-site read density, essentiality call	<i>E. coli</i>	1	901,383 insertion mutants	1 condition	9.0×10^5 †	9.0×10^5	K-12+transposon	A transposon mutant library estimated at approximately 3.7 million mutants, sequenced to 8279309 mapped reads over 901383 unique insertion sites, from which 358 genes were identified as essential. Only one condition was used, Luria broth at 37 degrees C to OD600 of 1.0, in two independent DNA extracts TL1 and TL2. 248 genes (59.5 percent) of the essential calls were common to this study, the Keio collection and PEC. The ENA accession is quoted from the paper’s data availability statement and the ENA page itself was not fetched.
12	Price 2018 Transposon fitness; barcode competitive fitness (BarSeq log ratio)	<i>E. coli</i>	1	3,789 insertion mutants	162 conditions	6.1×10^5 †	6.1×10^5	K-12+transposon	The <i>E. coli</i> subset is separable: the published per-organism page for orgId Keio (<i>Escherichia coli</i> BW25113) states 207 condition samples with 162 successful, and its fit_logratios_good.tab was downloaded and counted at 162 experiment columns by 3789 gene rows. Whole-study totals are 32 bacteria, 4870 genome-wide fitness experiments meeting consistency criteria, and 11779 poorly annotated protein-coding genes with phenotypes. A 2021 data correction removes the sucrose and D-mannitol experiments for this organism. The live Fitness Browser sits behind a Cloudflare challenge and could not be read, so its current counts may exceed these.
13	Choe 2025 CRISPR library screen; sgRNA abundance by next-generation sequencing	<i>E. coli</i>	2	39,591 guides	12 conditions	4.8×10^5 †	4.8×10^5	K-12+guide	The initial library holds 39591 sgRNA sequences at 99.96 percent coverage, targeting 4198 coding sequences. Twelve chemical conditions were evaluated: CCCP, polymyxin B, pyocyanin, rifampicin, sulfamethizole, verapamil, puromycin, erythromycin, phleomycin, mitomycin C, MMS and novobiocin. This is the genuine antibiotic-panel CRISPRi screen in <i>E. coli</i> and is the correct source for that claim rather than Rousset 2018. The ENA project was fetched and confirmed, released December 2024, submitted by KAIST.
14	Yunus 2026 Production campaign; isoprenol titer under predicted CRISPRi knockdowns	<i>P. putida</i>	4	20 guides	1 condition	60‡	60	KT2440+guide	Found while sweeping for CRISPRi isoprenol work and it clearly belongs: FluxRETAP and VAMMPIRE were used to pick downregulation targets and to assemble arrays of up to five sgRNAs, reaching nearly 1.5 g/L isoprenol by knocking down PP_4118 (alpha-ketoglutarate dehydrogenase). Bibliography verified through NCBI eutils. Strain and instance counts are estimates: the abstract does not state them and the full text is paywalled. No accession is given in the indexed metadata, so data_location is deliberately empty rather than guessed. Shares authors and the piY670-era pathway with Carruthers 2025 and Banerjee 2024, so genotype deduplication against those rows is required.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
15	de Siqueira 2025 Tolerance / robustness; growth rate on glucose-acetate mixtures plus isoprenol titer by GC-FID	<i>P. putida</i>	4	7 resequenced clones	3 conditions	63†	63	evolved-WGS	Both recalled accessions verified by fetching the repository records. PXD055153 is titled Restoration of growth and isoprenol production in dual carbon source media by an acetate-sensitive <i>Pseudomonas putida</i> strain, submitted 2024-08-23, linked to this DOI. PRJNA1153078 (<i>Pseudomonas putida</i> KT2440 tolerization on acetate and isoprenol) holds exactly 5 SRA experiments and 5 BioSamples, matching the 5 resequenced isolates, four tolerized Sigma-class strains plus the parental producer; 2 of the 7 characterized strains were not resequenced and are blocked at the genotype level. ONE study with linked assets PXD055153 plus PRJNA1153078. Proteome panel size is never stated numerically, so dimensionality is kept at 1 rather than guessed.
16	Wang 2015 isoprenol tolerance Tolerance / robustness; OD600 at 12 h with and without isoprenol, sample SD over 2 biological replicates	<i>E. coli</i>	2	47 released: wild type plus 46 deletions	2 isoprenol doses	94†	94	K-12-KO	The only isoprenol tolerance screen in either host over a cataloged deletion collection, which is what earns tier 2 on one cleared bar. Supplementary Table S3 releases OD600 with standard deviation for the wild type and 46 Keio single deletions at 0 and 0.5 percent isoprenol by volume, in 2YT at 30 degrees, and Table S4 adds a 9-gene transporter transcript panel. Keio JW numbers are given per strain, so the deletion alleles resolve to Baba 2006. Two multi-gene strains exist and are the paper's own epistasis test, <i>acrA</i> with <i>acrB</i> and that double with <i>tolC</i> , both reported as non-additive against their singles with a plasmid complementation control; neither appears in any released table, so the combinatorial arm is a provenance gap rather than a record. The 0.75 percent dose is figure-only for the same reason. The paper's mutant count is self-inconsistent, 44 in the results and 45 in the abstract against 46 rows in Tables S2 and S3; the table count is what is ingested.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
17	Lim 2025 isoprenol TALE Tolerance / robustness; maximum specific growth rate under isoprenol challenge plus isoprenol titer with the pIY670 plasmid	<i>P. putida</i>	4	46 resequenced clones	2 conditions	114†	114	evolved-WGS	The recalled design is correct and reported: WT, IPL300 and IPL400 each evolved in four independent lineages (12 isoprenol lineages), plus four extra WT lineages tolerized against formaldehyde as controls, so 16 lineages total, isoprenol ramped 4 to 8.5 g/L over 143 to 353 generations. Only the 46 whole-genome resequenced strains (16 endpoint clones plus about 30 intermediates) have a reconstructable genotype; the evolved populations themselves do not and are blocked at the genotype level. 158 unique mutations across 73 regions; proteomics quantified an average of 2,375 of 5,565 putative proteins. Linked assets: PXD054609 (confirmed, citing this PMID), GEO GSE281392, and ALEdb project Pputida_isoprenol_TALE. IMPORTANT: the stated resequencing accession PRJNA1187681 does NOT resolve at NCBI BioProject, NCBI SRA or ENA as of 2026-09-24, so the raw reads are not retrievable; the mutation calls survive only in Supplementary Dataset 1.
18	Tian 2019 isopentenol CRISPRi^B Combinatorial design; isopentenol titer by GC plus OD600, replicate count not stated	<i>E. coli</i>	4	24: 1 base, 18 single-guide, 3 two-guide, 2 three-guide	3 induction levels	84†	168	engineered-chassis+guide	The only isoprenol production campaign in either host with a nested combinatorial perturbation axis, and the reason this row is here despite releasing nothing. Three genes, asnA, gldA and prpE, are realized as 3 singles, all 3 pairs and the full triple, each assayed across repression levels of 0, 5 and 10 nM anhydrotetracycline, so the design scores an interaction and a dose rather than a ranking. A second 3-gene array over poxB, ackA and pta is matched against a triple knockout of the same genes, which is a knockdown-versus-deletion comparison on one gene set. The host is DH1, not BW25113 or MG1655, carrying the KG1R10 mevalonate pathway as two plasmids: atoB, HMGS, HMGR, MK and PMK under Ptrc, then nudB and PMD under PlacUV5. Every construct has a JBEI registry number. The paper's own target count is self-inconsistent, 21 genes in the introduction against 18 guides in the results and 15 in the abstract, and one printed symbol, arcC, is not a K-12 gene name.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
19	Kang 2026 isoprenyl acetate Production campaign; isoprenyl acetate and isoprenol titer from mixed glucose and xylose	<i>P. putida</i>	4	25 pathway designs	3 conditions	225‡	225	engineered-chassis	Found in a ProteomeXchange sweep rather than the recalled list, and it belongs only if the downstream ester is admitted: the product is isoprenyl acetate, an ATF1-esterified derivative of isoprenol, at 1.9 g/L in fed-batch with a yield of 0.067 g per g total sugar. PXD067010 confirmed by fetching its PRIDE record, which names this DOI and PMID and lists Petzold as lab head. Engineering combines a heterologous alcohol acetyltransferase, esterase deletions to stop product degradation, glucose-xylose co-utilization, and reinforced acetyl-CoA flux, across tube, flask and 2 L fed-batch in biological triplicate. Strain and instance counts are estimates; the deposit does not state a run count. Shares Carruthers and Lee with the flagship isoprenol rows, so deduplicate genotypes.
20	Menasalvas 2025 Production campaign; isoprenol titer by GC-FID after biosensor-coupled pyrF growth selection	<i>P. putida</i>	3	165 deletions	2 conditions	1.3×10^3 ‡	1.3×10^3	engineered-chassis	ATtribution correction: no author named Teng is on this paper. The Dryad DOI belongs to Menasalvas et al. with Thomas Eng as co-corresponding author, so the recalled Teng is almost certainly Eng. Over 165 deletion and overexpression strains across 70 gene loci, quadruplicate, titers at 24 and 48 h, reaching about 900 mg/L (36-fold). ONE study, four linked deposits, all confirmed: Dryad 10.5061/dryad.sbccc2frjq (2 files, 78.98 MB), PRIDE PXD061547 (An Isoprenol Biosensor for Combinatorial High Throughput Strain Engineering in Pseudomonas putida, 97 raw files with 8, 24 and 48 h timepoints), BioProject PRJNA1226229 (3 WGS runs), Zenodo 10.5281/zenodo.17155686 (AlphaFold output only). IMPORTANT on the CRISPRi library question: the paper does contain a pooled dCpf1 library of about 16,500 guides (3 per gene over 5,591 coding sequences), but there is NO released per-guide fitness or abundance matrix. Dryad holds only baseline per-guide read counts, a lost-guide list, per-round enriched-guide hit lists, and representative rather than per-replicate ONT reads, so a guide-by-sample matrix cannot be reconstructed. Proteomics covers about 2,500 to 3,000 proteins on a five-context subset, not every strain, so dimensionality is kept at 1.
End of tranche 1. Rows 1–20 are the first builds, 6 per host.									

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
21	Rapp 2026 CRISPRi metabolome CRISPR library screen; flow-injection MS metabolome (ion intensities), plus carotenoid titer in the engineering follow-up	<i>E. coli</i>	3	1,515 guides	1 condition	1.5×10^3	4.5×10^5	K-12+guide	Arrayed CRISPRi library covering all 1515 metabolic genes of iML1515, each strain profiled by fast flow-injection MS, so per-design metabolome vectors are recoverable; 36% of iML1515-predicted metabolites accumulated specifically in some knockdown, and LC-MS/MS spectra were generated for 102 previously uncharacterized metabolites. The exact number of annotated ions per strain was not verified from the full text, so dimensionality 300 is an estimate. The MassIVE record was fetched and confirmed.
22	Schmidt 2022 nitrogen Transposon fitness; gene-level RB-TnSeq fitness across nitrogen-containing sole nitrogen sources	<i>P. putida</i>	1	4,778 insertion mutants	71 conditions	3.4×10^5 †	3.4×10^5	KT2440+transposon	Not on the requested list but it clearly belongs, and it is the largest single-study condition axis for this organism: assimilation of 52 different nitrogen-containing compounds across 71 tested conditions, with significant fitness phenotypes in 672 genes including 100 transcriptional regulators and 112 transport proteins. Same JBEI-1 library and same Deutschbauer and Keasling pipeline as the lysine, valerolactam and fatty-acid rows, and it is one of the source studies cited by the Borchert 2024 compendium, so it must be deduplicated against that row. The compendium's nitrogen-source experiment group holds 104 of the 332 samples, which is where most of these land. Fitness Browser only, Cloudflare-blocked, so unconfirmed.
23	Wang 2018 CRISPR library screen; sgRNA abundance by Illumina sequencing, gene and sgRNA fitness	<i>E. coli</i>	2	56,071 guides	5 conditions	2.8×10^5 †	2.8×10^5	K-12+guide	Two corrections to the commonly cited form of this row. The venue is Nature Communications 9:2475, not Science; no Science 2018 genome-wide <i>E. coli</i> CRISPRi paper could be found. The library is 55671 targeting sgRNAs plus 400 negative controls, not about 92000; the 92000-guide figure belongs to the Bikard-lab libraries of Cui 2018 and Rousset 2018. Coverage is at least one sgRNA for 98.6 percent of 4140 protein-coding genes and 79.8 percent of 178 RNA-coding genes. The five screened phenotypes are essentiality in LB, auxotrophy in MOPS versus LB, L-tryptophan biosynthesis, furfural tolerance and isobutanol tolerance. Processed per-gene and per-sgRNA fitness are released as supplementary data; no raw-sequencing accession could be confirmed.
24	Rousset 2018 CRISPR library screen; guide abundance log2 fold change (relative sgRNA fitness)	<i>E. coli</i>	2	59,000 guides	4 conditions	2.4×10^5 †	2.4×10^5	K-12+guide	The screen starts from a pool of about 92000 sgRNAs targeting random chromosomal positions and is filtered to about 59000 guides. Conditions are growth in rich medium over 17 generations for essentiality plus infection by phages lambda, T4 and 186 at MOI 1, all in triplicate; the paper screens no antibiotics, so any title citing antibiotic exposure for this paper is a conflation. 379 genes were selected for follow-up, including 235 annotated as essential. The ENA accession was verified directly at EBI.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
25	Shiver 2016 Fitness / chemical genomics; colony opacity fitness score (Iris)	<i>E. coli</i>	1	3,975 deletions	57 conditions	2.3×10^5 †	2.3×10^5	K-12-KO	3975 Keio deletion mutants against 26 new stresses for 57 conditions in total, scored from colony opacity with the Iris image analysis software, and designed as an extension of the Nichols 2011 condition set. The release is raw plate images plus Iris output and ranked score files rather than a clean gene by condition matrix, so building the matrix requires rerunning the published scoring scripts. The Zenodo mirror of the Dryad deposit was fetched and its file list confirmed.
26	Thompson 2020 fatty acid and alcohol Transposon fitness; gene-level RB-TnSeq fitness on fatty acid and alcohol carbon sources	<i>P. putida</i>	1	4,778 insertion mutants	23 conditions	2.2×10^5 †	2.2×10^5	KT2440+ transposon	The recalled 13 fatty acids and 10 alcohols is confirmed verbatim in the abstract and the compounds were enumerated: straight-chain C3 to C10 plus C12 and C14, the esters Tween 20 and butyl stearate, oleic acid; and ethanol, butanol, pentanol, 1,2-propanediol, 1,3-butanediol, 1,4-butanediol, 1,5-pentanediol, isopentanol, isoprenol, 2-methyl-1-butanol. Biological duplicate throughout, corroborated by the compendium metadata showing exactly 2 samples per compound, hence $23 \times 2 \times 4,778$; the condition-collapsed count is about 109,900. Directly relevant to the isoprenol rows: this paper postulates the catabolic routes for isoprenol and isopentanol through leucine metabolism after oxidation and CoA activation, which is the product-loss pathway the isoprenol producers delete. A published correction exists (AEM 87(8):e00177-21, DOI 10.1128/aem.00177-21). Fitness Browser only, Cloudflare-blocked, so unconfirmed; not every published condition survived the compendium’s completeness filter, so the full matrix is not recoverable from the workbook alone.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
27	Borchert 2023 lignin tolerance Transposon fitness; gene-level RB-TnSeq fitness under lignin-stream inhibitors on a fixed glucose background	<i>P. putida</i>	1	4,732 insertion mutants	16 conditions	2.1×10^5 †	2.1×10^5	KT2440+transposon	This is the lignin-stream tolerance row with reconstructed engineering strategies, and it is a tolerance rather than a catabolism screen: every stressor is overlaid on M9 plus 20 mM glucose. The 15 stressors were enumerated from the SRA sample titles (ferulate, 4-coumarate, levulinate, beta-ketoadipate, protocatechuate, cis,cis-muconate, glycolate, lactate, acetate, Na2SO4, NaCl, vanillate, vanillin, 4-hydroxybenzoate, 4-hydroxybenzaldehyde) plus a glucose-only control, in triplicate: the 57 SRA runs decompose exactly as 15 x 3 selective, plus 3 plus 3 glucose-control enrichments, plus 3 plus 3 time-zero samples. PRJNA856070 confirmed by fetching the BioProject page (57 SRA experiments, 57 BioSamples). Reconstructed strategies from the fitness hits: delta-gacAS, delta-fleQ, delta-lapAB, delta-ttgR with Ptac-ttgABC, Ptac-PP_1150-PP_1152, delta-relA, delta-PP_1430, several of which also improved growth in a complex lignin-stream mimic. These samples are NOT in the Fitness Browser, which is precisely why the Borchert 2024 compendium reached into SRA for them; the 39 samples of Putida_ML5_set100 are recoverable at full precision from that workbook.
28	Schastnaya 2021 Metabolome / flux; metabolome of phosphosite point mutants and matched deletion mutants	<i>E. coli</i>	4	89 designs	5 conditions	445†	2.0×10^5	engineered-chassis	52 known phosphosites on 23 central metabolic enzymes were mutated in E. coli MG1655 delta-mutS, giving 89 phosphomutants profiled on 5 carbon sources against matched Keio deletion strains, with 460 to 500 annotated metabolites by flow-injection TOF-MS. This is the best genotype-by-condition phospho-regulation panel, but it generated no phosphoproteome of its own; its phosphosite list is drawn from earlier studies such as Potel 2018.
29	Cui 2018 CRISPR library screen; guide abundance log2 fold change over 17 generations	<i>E. coli</i>	2	92,000 guides	2 conditions	1.8×10^5 †	1.8×10^5	K-12+guide	The library is about 92000 unique guide RNAs targeting random positions along the MG1655 genome with an NGG PAM requirement, averaging 19 targets per gene; an exact targeted-gene count is not stated. The two genome-wide screens differ only in dCas9 expression level (strains LC-E18 and LC-E75), both in rich medium with anhydrotetracycline in triplicate, with no stress panel. Data availability names Supplementary Data 4 for the screen results and otherwise defers to the corresponding author on request, so there is no SRA, ENA or GEO accession for this paper.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
30	Schmidt 2016 Proteome; absolute protein copies per cell and per cell volume	<i>E. coli</i>	4	3 wild type	22 conditions	66†	1.6×10^5	reference-only	2,359 proteins quantified across all 22 conditions, about 55 percent of predicted ORFs, calibrated to copies per cell with flow cytometry and synthetic peptide standards. Primary strain BW25113, with MG1655 and NCM3722 at selected conditions; biological triplicates for the main dataset, so 66 is a product not a reported sample count.
31	Butland 2008^B Genetic interaction; colony size interaction (S) score	<i>E. coli</i>	1	155,415 double mutants	1 condition	1.6×10^5 ‡	1.6×10^5	K-12-KO x KO	Only the citation fields and the abstract-level method description could be confirmed: a quantitative screening procedure based on conjugation of <i>E. coli</i> deletion or hypomorphic strains to create double mutants on a genome-wide scale. The article is closed access with no PMC deposit, and nature.com, Springer static content and mirror sites all refused fetching, so the number of query genes, array strains and screened pairs are all unconfirmed. A search snippet suggesting 39 genome-wide screens, which would give roughly 39 by 3985 or 155415 pairs, could not be verified on any fetched page, so the counts above are an unverified estimate and the row is marked blocked pending a paywalled retrieval.
32	Mori 2021 Proteome; absolute protein mass fractions by DIA/SWATH with xTop inference	<i>E. coli</i>	4	3 wild type	60 conditions	66	1.5×10^5	reference-only	2,335 proteins with absolute mass fractions over about 60 growth conditions in 66 samples, covering carbon, nitrogen, phosphate and oxygen limitation, non-metabolic stresses and biofilm states, in K-12 NCM3722 and derivatives plus MG1655 sub-strain EQ353 and <i>E. coli</i> Nissle 1917. No ProteomeXchange or PRIDE accession was stated in the text examined; the deposit found is the SWATHAtlas spectral library. An author correction was published in October 2024 and should be read with the paper.
33	Ishii 2007 Multi-omics campaign; paired transcriptome, proteome, metabolome and 13C flux in glucose-limited chemostats	<i>E. coli</i>	4	25 designs	5 conditions	29	1.2×10^5	K-12-KO	Design confirmed from a secondary open-access source: 24 single-gene disruptants at a fixed dilution rate of 0.2 per hour plus wild type at 5 dilution rates, measured by DNA microarray, 2D-DIGE, CE-TOFMS and metabolic flux analysis. Per-layer depth (transcripts, proteins, metabolites, fluxes) is NOT confirmed: the paper is paywalled and absent from PMC, and no accessible secondary source quotes those counts, so 4,300 is a genome-wide-array estimate for the dominant layer and must be replaced before use in any ranking.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
34	Campos 2018 Fitness / chemical genomics; single-cell microscopy morphology and cell cycle features	<i>E. coli</i>	2	4,227 deletions	1 condition	4.2×10^3 †	1.1×10^5	K-12-KO	4227 strains of the Keio collection, covering 98 percent of the non-essential genome and 87 percent of the complete genome, imaged in one condition (M9 with 0.1 percent casamino acids and 0.2 percent glucose at 30 degrees C). 26 quantitative features per strain: 19 morphological, 5 cell cycle, and 2 growth-related. On average about 360 cells were imaged per strain, roughly 1.3 million cells after filtering. This is the highest-dimensionality <i>E. coli</i> deletion phenotype release found, and the natural counterpart to a yeast morphology or transcriptome profile rather than to a scalar fitness score.
35	Typas 2008^B Genetic interaction; colony pixel count interaction score	<i>E. coli</i>	1	91,655 double mutants	2 conditions	9.2×10^4 †	9.2×10^4	K-12-KO x KO	This is a method paper (GIANT-coli). Its released per-pair data cover only two demonstration screens: a 12 by 12 cross yielding 66 distinct pairwise double mutants plus 12 self-matings, and the pal and yraP screens in Supplementary Tables 2A and 2B. The genome-wide work is reported only as a linkage curve from 9 by 3985 crosses in LB and 14 by 3985 in M9, which is the 91655 figure above and is a product, not a released matrix. Status is blocked because no genome-wide double-mutant score matrix was released. The toolkit couples the Keio kanamycin deletion library with the ASKA chloramphenicol library, about 4000 single-gene deletions each.
36	Girgis 2009 Transposon fitness; microarray genetic footprinting z-score per locus	<i>E. coli</i>	1	500,000 insertion mutants	17 conditions	7.3×10^4 ‡	7.3×10^4	K-12+ transposon	The library is about 5 by 10^5 mutants each with a single transposon insertion, selected in 17 antibiotics, read out by hybridizing transposon-junction DNA against genomic DNA on spotted microarrays. The released data are per-locus z-scores, with Dataset S5 the combined z-scores across all loci, so usable records are locus by drug rather than strain by drug; the number of array loci is not stated, so the instance count here is an estimate from an assumed roughly 4300 loci and is labeled as such. No GEO or ArrayExpress accession exists for this dataset; a GEO search returned only the unrelated expression series GSE10855. The companion motility screen is Girgis HS, Liu Y, Ryu WS, Tavazoie S 2007 PLoS Genetics 3(9):e154, doi 10.1371/journal.pgen.0030154, PMID 17941710, same library size, four selection regimes released as 13 supplementary microarray datasets with no accession.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
37	Banerjee 2025 Production campaign; indigoidine titer from p-coumarate plus growth rate, with a global DIA proteome	<i>P. putida</i>	4	12 promoter variants	3 conditions	36‡	7.2×10^4	KT2440+promoter	This is the recalled genome-scale design-tradeoff campaign, and it is the cleanest promoter-variant genotype axis I found for KT2440: a four-gene growth-coupling cutset for p-coumarate to glutamine to indigoidine, whose fourth gene PP_0897 (a fumarate hydratase isomer) proved multifunctional and rate-limiting, so its native promoter was swapped for the Anderson-collection pJ23109 or the weak PP_0415 promoter. PXD050285 verified two ways (ProteomeCentral and the PRIDE v3 API) and its files enumerated: 26 raw MS runs, Orbitrap Exploris 480, DIA searched library-free with DIA-NN at 1 percent global FDR, comprising an 8-run promoter-variant arm (pJ and 0415 driving PP_0897, 4 replicates each) and an 18-run cross-feeding arm (strains 2370 and 2487 across M9 alanine, M9 alanine malate, M9 p-CA alanine malate, n=3). The about 2,000 quantified proteins is REPORTED verbatim in the paper, not estimated. A separate 190-substrate BIOLOG grid in the same paper is a much larger environment axis if it is ingested. Note the PRIDE submission title differs from the published title; both are correct for their object.
38	Fang 2025 FFA CRISPRi-FACS^B CRISPR library screen; sgRNA enrichment (FACS sort plus NGS), then titer (g/L) on validated strains	<i>E. coli</i>	2	55,671 guides	1 condition	5.6×10^4	5.6×10^4	K-12+guide	The screen reuses a previously published plasmid library of 55671 sgRNAs, sorted with a fluorescent free-fatty-acid biosensor and read out by NGS; the engineered pcnBi-acrDi-fadR+ strain reached 35.1 g/L FFAs in fed-batch. Pooled enrichment gives a guide-level score rather than a per-design titer, so only the handful of individually reconstructed strains carry phenotype values; no sequencing accession was confirmed.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
39	Babu 2014 Genetic interaction; colony size epistasis (S) score	<i>E. coli</i>	1	671,071 double mutants	1 condition	4.3×10^4	4.3×10^4	K-12-KO x KO	The largest <i>E. coli</i> digenic screen that could be verified. It reports over 600000 digenic mutant combinations from 163 query donor genes crossed into an array of 3968 non-essential deletions plus 149 hypomorphic strains, which is 671071 nominal pairs, with donors transferred by conjugation from an Hfr-Cavalli chloramphenicol-marked background. Only the high-confidence subset is released, 25239 aggravating plus 17466 alleviating for 42705 pairs in Table S2, so the full matrix is not recoverable and the instance count reflects the released subset. The dedicated portal http://ecoli.med.utoronto.ca/esga no longer resolves to the Emili lab (the host presents a certificate for lymnaea.org), and BioGRID has no entry for PMID 24586182. The earlier Babu 2011 PLoS Genetics map (doi 10.1371/journal.pgen.1002377, PMID 22125496) is smaller at over 235000 combinations and likewise released only significant pairs. Kumar 2016 Cell Reports (doi 10.1016/j.celrep.2015.12.060, PMID 26774489) covers 102255 viable digenic mutants and is the one <i>E. coli</i> interaction set BioGRID serves, at 31644 curated interactions.
40	Gupta 2024 turnover Translation / turnover; per-protein degradation and turnover rates	<i>E. coli</i>	3	1 wild type	13 conditions	13†	4.2×10^4	reference-only	About 3,200 proteins, 77 percent of all genes, with turnover rates across 13 growth conditions in strain NCM3722, by heavy ^{15}N dynamic labeling with TMTproC complement-reporter mass spectrometry. Replicates exist but the per-condition replicate count is unconfirmed, so instances is a floor. Cite the 2024 Nature Communications version, not the 2022 bioRxiv preprint.
41	MCF2Chem 2023^A Aggregation / support; titer, yield, productivity, content per literature record	<i>E. coli</i>	3	8,888 designs	1 condition	8.9×10^3	3.6×10^4	reference-only	8888 production records covering 1231 compounds in 590 microbial cell factories, with titer, yield, productivity and content plus strain culture information; bacteria are 60% of species, and <i>E. coli</i> , <i>S. cerevisiae</i> , <i>Y. lipolytica</i> and <i>C. glutamicum</i> together account for 78% of the compounds, with <i>E. coli</i> alone producing about a quarter of the compounds among the top 20 species. No exact <i>E. coli</i> record count is published, so the <i>E. coli</i> subset size could not be determined. These are counts of literature records, not measured strain-condition observations, and must be resolved to primary source DOIs before they count as independent datasets.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
42	CeCaFDB flux compendium ^A Aggregation / support; curated published 13C flux distributions on a normalized central-carbon network	<i>E. coli</i>	3	mixed	not stated	297	2.3×10^4	reference-only	581 flux distributions across 36 organisms from 118 references (1995 to 2013); <i>E. coli</i> alone contributes 297 flux distributions from 32 references, the largest single-organism set (<i>S. cerevisiae</i> is next at 76). The normalized comparison network has 66 metabolites and 76 reactions, which is the per-map dimensionality bound, not necessarily each source study's own network size. The site returned an expired-TLS error on 2026-09-24, so current operation is unconfirmed.
43	Thompson 2019 lysine Transposon fitness; gene-level RB-TnSeq fitness on glucose, 5-aminovalerate, D-lysine and L-lysine	<i>P. putida</i>	1	4,778 insertion mutants	4 conditions	1.9×10^4 †	1.9×10^4	KT2440+ transposon	This is where the name JBEI-1 is defined, verbatim: the JBEI-1 library was created by diluting a 1 mL aliquot of the previously described <i>P. putida</i> RB-TnSeq library (Rand 2017) into 500 mL of LB plus kanamycin. So JBEI-1 is a regrown working stock of Putida_ML5, not a separate library, which is why every transposon row here shares one join key. Four sole-carbon conditions at 10 mM in MOPS minimal medium, 48 h, with three time-zero aliquots; Methods state no per-condition replicate count. The paper reports 39 genes with fitness below -2 at absolute t above 4. Fitness data are said to be publicly available at fit.genomics.lbl.gov with no per-experiment accession, and that site returns a Cloudflare challenge, so unconfirmed. Strains and plasmids at public-registry.jbei.org folder 391.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
44	Rand 2017 Putida__ML5 library Modality / backbone; defines the barcoded insertion pool every fitness row is scored on; itself assayed on four carbon sources	<i>P. putida</i>	3	185,401 insertion mutants	4 conditions	1.9×10^4 †	1.9×10^4	KT2440+ transposon	CORRECTION to the recalled library definition on two points. First, the library is named Putida__ML5 and was built here in the Deutschbauer and Arkin lab from the pKMW3 mariner vector library of Wetmore 2015; JBEI-1 is not a separate library but a regrown, re-aliquoted working stock of Putida__ML5 named later in Thompson 2019 mBio. The compendium metadata’s mutantLibrary column carries exactly two values, Putida__ML5 (25 samples) and Putida__ML5_JBEI (307 samples). Second, the recalled sizes of about 100,000 insertion mutants and about 4,800 nonessential genes are BOTH wrong. The only quantitative description in the literature is in Borchert 2022 (ACS Synth Biol 11:2015-2021, DOI 10.1021/acssynbio.2c00119, PMID 35657709) citing this paper: 185,401 uniquely barcoded transposon insertions, of which 32,591 are intergenic and 152,810 map to 5,213 of 5,661 genes. The about 4,800 figure is a FITNESS-coverage number, not library coverage: the Fitness Browser states fitness data for 4,778 of 5,661 genes, and Borchert 2024 reports 4,732 of 5,564 after filtering. Rand 2017 itself says only thousands of kanamycin-resistant colonies. Its own screen covers 4 carbon sources in duplicate over two days. Join on barcode to mapped insertion to PP__ locus tag against AE015451 (6,181,873 nt, taxid 160488); no study releases per-barcode fitness, all publish gene-level aggregates. The Fitness Browser organism page is orgId=Putida and reported 314 experiments across 54 carbon sources, 52 nitrogen sources, 2 stress compounds and 27 other conditions, but it is behind a Cloudflare managed JavaScript challenge today, so those figures were read from Internet Archive snapshots (2025-03-10 and 2025-04-24) and must NOT be recorded as live 2026 values. Hence accession_confirmed false.
45	Balakrishnan 2022 Multi-omics campaign; matched mRNA and protein mass fractions plus promoter activity and mRNA degradation rates	<i>E. coli</i>	4	2 designs	4 conditions	8†	1.5×10^4	reference-only	The cleanest transcript-and-protein join on identical samples: more than 1,900 proteins with matched mRNA, plus about 2,700 genome-wide mRNA degradation rates, in <i>E. coli</i> K-12 NCM3722 (and a delta-rsd mutant) across a reference glucose minimal medium plus carbon limitation, anabolic limitation and translational inhibition, growth rates 0.3 to 0.9 per hour. Replicate counts vary by assay and are not uniformly reported, so instances is a product.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
46	Brunk 2016 Multi-omics campaign; 86 metabolites plus 55 protein complexes, with product titers (isopentenol, limonene, bisabolene)	<i>E. coli</i>	4	9 pathway designs	9 conditions	81†	1.1×10^4	engineered-chassis	Nine strains (eight engineered plus wild-type <i>E. coli</i> DH1) carrying three versions of a heterologous mevalonate pathway, sampled across a 0 to 72 hour batch fermentation: the joint panel is 86 metabolites and 55 protein complexes at 9 time points, while the aggregate metabolomics set is described as 9 strains x 13 time points x 86 metabolites, so instances 81 is the product for the joint panel and the metabolome alone is larger. Variances come from triplicate measurements where available. No omics repository accession was confirmed.
47	Thompson 2019 valerolactam Transposon fitness; gene-level RB-TnSeq fitness on lactam carbon sources, then valerolactam titer in deletion strains	<i>P. putida</i>	2	4,778 insertion mutants	2 conditions	9.6×10^3 †	9.6×10^3	KT2440+transposon	Two selective RB-TnSeq conditions, 10 mM valerolactam and 10 mM 5-aminovalerate in MOPS minimal medium, with three time-zero aliquots; the two 500 uL aliquots per condition are pooling, not replication, so the paper reports no replicate count. RB-TnSeq is the small half of this paper; the payload worth ingesting is the engineering ladder it produced, wild type 0 mg/L to delta-oplBA 9.27 to delta-oplBA delta-davT 85.19 to delta-oplBA delta-davT delta-alr 91.97 mg/L valerolactam at 48 h, with strains at public-registry.jbei.org folder 456. Fitness lives only in the Fitness Browser with no per-experiment accession, and that site is behind a Cloudflare managed challenge to both WebFetch and curl, so accession_confirmed is false. The two valerolactam carbon-source samples are however confirmed to exist, appearing by name (set6IT064, set7IT045) in the Borchert 2024 compendium workbook.
48	D2Cell 2026 ^A Aggregation / support; literature-reported production metric per gene-modification record	<i>E. coli</i>	3	8,134 designs	1 condition	8.1×10^3	8.1×10^3	reference-only	Confirmed verbatim in the preprint: 8134 experimentally reported single- and double-gene modification records for 73 products in <i>E. coli</i> , merged for training with 11643 single-gene-modification rows for the same 73 products simulated by the genome-scale model (FSEOF-style), so the simulated rows originate from a separate generation step and are separable in principle, but the preprint states no explicit flag distinguishing them in the released file; the whole database is 29006 entries, 1210 products, 751 organisms. These are counts of literature records, not measured strain-condition observations, and must be resolved to primary source DOIs before they count as independent datasets.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
49	Li 2014 Translation / turnover; absolute protein synthesis rates from ribosome profiling	<i>E. coli</i>	3	1 wild type	2 conditions	2†	6.1×10^3	reference-only	3,041 genes accounting for more than 96 percent of total protein synthesized in rich defined medium, with a similar number in glucose minimal medium, at 90 million fragments per sample. Method is ribosome profiling calibrated to total protein per doubling; the quantitative mass spectrometry comparison is to published external datasets, not generated here. Strain designation is unconfirmed (main-text methods do not name it).
50	Potel 2018 phosphoproteome Proteome; phosphosite identifications including phosphohistidine	<i>E. coli</i>	4	1 wild type	2 conditions	2‡	4.3×10^3	reference-only	The deepest <i>E. coli</i> phosphoproteome found, but with almost no breadth: one wild-type MG1655 strain on glycerol, exponential versus stationary phase. The 2,129 total phosphosites and 246 phosphohistidine sites on 173 proteins come from a secondary search snippet, not from a fetched primary text; the abstract confirms only that about 10 percent of sites are phosphohistidine, so dimensionality needs primary confirmation before ranking.
End of tranche 2. Rows 21–50 complete the fifty; rows below are the ranked reserve.									
51	Baba 2006 Modality / backbone; deletion viability (in-frame knockout obtained or not)	<i>E. coli</i>	3	3,985 deletions	1 condition	4.0×10^3	4.0×10^3	K-12-KO	Of 4288 genes targeted, mutants were obtained for 3985, held in duplicate for 7970 strains; 303 genes including 37 of unknown function could not be disrupted and are the paper’s essential-gene candidates. Strain background is <i>E. coli</i> K-12 BW25113. The distribution URL given in the paper (GenoBase, http://ecoli.aist-nara.ac.jp/) is stale; NBRP <i>E. coli</i> currently lists 3909 Keio entries ready to distribute. This is the strain resource and the join axis for Nichols 2011, Tong 2020, Campos 2018, Shiver 2016 and the conjugation-based interaction screens, not a phenotype screen itself.
52	Gerdes 2003 Transposon fitness; genetic footprinting essentiality assertion	<i>E. coli</i>	2	3,746 insertion mutants	1 condition	3.7×10^3	3.7×10^3	K-12+ transposon	Unambiguous essentiality assessments were made for 3746 (87 percent) of <i>E. coli</i> protein-encoding genes or ORFs, of which 620 (14 percent) were asserted essential and 3126 (73 percent) non-essential; no assertion was possible for 327 genes on technical grounds and evidence was insufficient for 218. Method is Tn5-based genetic footprinting in rich aerobic media. The per-gene Table S1 is still served live at the Wisconsin mirror, which makes this the cleanest hash-pinnable per-gene essentiality table for MG1655. Note that the Database of Essential Genes lists 609 essential genes for this study and 296 for Baba 2006, both of which disagree with the source papers (620 and 303); use the paper tables.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
53	Kochanowski 2017 Multi-omics campaign; promoter activities and absolute central metabolite concentrations on shared conditions	<i>E. coli</i>	4	2 wild type	26 conditions	26	3.7×10^3	reference-only	95 fluorescent promoter reporters (an existing library expanded by 28, with 31 inactive promoters discarded) measured at steady state across 26 conditions spanning carbon sources, amino-acid supplementation and sub-lethal chloramphenicol, growth rates 0.1 to 1.5 per hour, in BW25113 plus a Crp deletion mutant. 47 central metabolites were quantified absolutely by ion-pairing UPLC-MS/MS in 23 of the 26 conditions, so dimensionality 142 is the union of 95 promoter activities and 47 metabolites, not a per-sample vector measured in every condition. No fluxes were measured.
54	Oyetunde 2019^A Aggregation / support; titer (g/L), rate (g/L/h), yield (g/g)	<i>E. coli</i>	3	1,200 designs	1 condition	1.2×10^3	3.6×10^3	reference-only	About 1200 experimentally realized <i>E. coli</i> cell factories hand-curated from about 100 papers covering more than 20 products, with features in six categories (carbon source, bioprocess conditions, genetic modifications, product properties, production metrics, undocumented factors) and TRY targets; data availability statement says all data are in the supplementary files. These are literature records, not measured strain-condition observations, so they must be resolved to primary source DOIs before counting as independent datasets.
55	Taniguchi 2010 Multi-omics campaign; single-cell protein copy number distribution and mRNA copy number per tagged gene	<i>E. coli</i>	3	1,018 designs	1 condition	1.0×10^3	2.0×10^3	engineered-chassis	1,018 confirmed strains of 1,400 attempted, each carrying a chromosomal C-terminal YFP fusion to one native gene at its native locus, imaged at about 4,000 cells per strain (96 strains per microfluidic device, 25 seconds each), with mRNA by RNA-seq and single-molecule FISH (137 genes had matched FISH). Per-strain dimensionality is only 2 summary values, so its value is the 1,018-gene tagged panel and the single-cell distributions, not depth per sample. The CGSC availability claim is unverified.
56	Haverkorn van Rijsewijk 2011 Metabolome / flux; 13C metabolic flux ratios in central carbon metabolism	<i>E. coli</i>	4	91 deletions	2 conditions	182†	1.5×10^3	K-12-KO	91 regulator deletion mutants (81 transcription factors plus 10 sigma and anti-sigma factors) on glucose and galactose, giving 8 metabolic flux ratios per strain by GC-MS of proteinogenic amino acids with FiatFlux. Dimensionality is genuinely small; the value of this row is the 91-strain deletion axis, not depth. Do not confuse with the Nielsen commentary (DOI 10.1038/msb.2011.10, PMID 21451588).

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
57	Opgenorth 2019 Multi-omics campaign; 1-dodecanol titer (g/L) plus targeted proteomics of the engineered pathway proteins	<i>E. coli</i>	4	60 RBS variants	2 conditions	180‡	900	engineered-chassis+RBS	Two DBTL cycles over 60 engineered <i>E. coli</i> MG1655 strains; cycle 1 tested 36 strains varying ribosome-binding sites and reductase enzymes in one pathway operon, and cycle 2 designs from the machine-learning models raised titer 21% to 0.83 g/L. The abstract states dodecanol plus all engineered-pathway protein concentrations were measured; the exact proteomics panel size and replicate count were not verified because the full text is paywalled, so dimensionality 5 and instances 180 are estimates. Conditions counted as 2 for the two DBTL cycles.
58	Valencia 2022 free fatty acids Production campaign; free fatty acid and fatty acid methyl ester chain-length distribution at 48 h	<i>P. putida</i>	4	48 pathway designs	3 conditions	144‡	864	engineered-chassis	Included because its readout is genuinely multivariate: every sample yields a chain-length vector (C8, C10, C12:1, C14, C16, C16:1) rather than a scalar titer, so encoding it at dimensionality 1 would discard the study. Eight host backgrounds from deletions of three fatty acyl-CoA ligases, crossed with about six plasmid constructs carrying protein-engineered TesA thioesterase variants or MmFAMT, across three media including fourfold-diluted sorghum hydrolysate, n=3; maxima 670.9 mg/L total free fatty acid, 253.6 mg/L C8, 302.4 mg/L total methyl ester, 561.1 mg/L on hydrolysate. The genotype count is an estimate because the host by plasmid design is not fully crossed. Source data for Figures 2 to 6 sit in Supplementary Data 1, but data availability cites only the bare public-registry.jbei.org root with no folder number, so nothing is retrievable by accession.
59	Mohamed 2020 hydroxycinnamic TALE Tolerance / robustness; growth rate and lag phase on p-coumarate and ferulate at increasing concentrations	<i>P. putida</i>	4	105 resequenced clones	4 conditions	420‡	420	evolved-WGS	Not on the recalled list but it is the best deposited <i>P. putida</i> tolerance evolution campaign. Six independent parallel replicates each for p-coumarate, ferulate and an equal-mass mixture, plus four replicates for a glucose control ALE, ramped to the solubility limit (40 g/L ferulate) or a 67-day cap; outcomes include a 37 h decrease in lag phase at 20 g/L p-coumarate and a 2.4-fold growth-rate increase at 30 g/L ferulate. PRJNA624660 confirmed by fetching the BioProject page: 102 SRA experiments and 105 BioSamples, with the ALEdb project named <i>P. putida</i> Hydroxycinnamic TALE. CAVEAT on genotype reconstruction: the deposit mixes endpoint populations with clonal isolates (1 to 2 intermediate isolates, the endpoint population, and a single endpoint isolate per condition), and only the isolates have a reconstructable single genotype; the population samples are blocked at the genotype level even though their reads are public.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
60	Fang 2021 FFA CRISPRi CRISPR library screen; free fatty acid titer (g/L) by GC	<i>E. coli</i>	4	108 guides	1 condition	324‡	324	K-12+guide	Arrayed rather than pooled: 108 synthetic sgRNAs repressing 108 chromosomal genes across five metabolic modules, each strain measured for FFA titer by GC in three biological replicates, giving recoverable per-design phenotypes; 30 beneficial genes from the screen plus 26 more from omics, 56 total, and 30.0 g/L in fed-batch. Accessions are named in the paper but were not fetched.
61	Jones 2015 ePathOptimize Combinatorial design; violacein titer (mg/L) by HPLC	<i>E. coli</i>	4	307 promoter variants	1 condition	307	307	engineered-chassis+promoter	Five IPTG-inducible mutant T7 promoters of graded strength were placed on each of the five violacein pathway genes vioABCDE in <i>E. coli</i> BL21star(DE3), a theoretical 3125-member library; 107 colonies were screened first (3.4% of the library) and a second enriched library of 200 mutants was screened, giving 307 genotype-linked titers. Best screen hit 238 mg/L, 63-fold over control, rising to 1829 plus or minus 46 mg/L after fermentation optimization. Deoxyviolacein and pathway intermediates were also quantified for selected strains, so the per-instance panel may exceed 1.
62	Jervis 2019 Combinatorial design; limonene titer (mg per L organic phase)	<i>E. coli</i>	4	88 RBS variants	1 condition	264‡	264	engineered-chassis+RBS	Two RBS libraries in <i>E. coli</i> DH10-beta: pGLlib pairs 12 designed RBS variants each for trAg-gpps and trMs-limS, 144 possible combinations, from which 360 colonies were screened in deep-well plates and 64 clones sequenced, covering 56 of the 144 combinations; a second 5184-member pMVA2 library had 156 colonies screened and a designed 32-combination reduced library built and tested. Genotypes 88 counts only sequence-resolved designs (56 plus 32), since unsequenced colonies have titers without a recoverable genotype; instances assumes triplicate.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
63	Schmidt 2025 3-hydroxyacid PKS Production campaign; LC-MS titer of three 3-hydroxyacids in supernatant at 48 h	<i>P. putida</i>	4	12 pathway designs	2 conditions	72‡	216	engineered-chassis	This is the modular type I PKS row. JBEI public registry folder 887 (Schmidt et al 2025) was confirmed through the REST endpoint: 58 STRAIN entries with public read access, which is an upper bound on strains built and the reason the genotype count is a figure-level estimate rather than a stated total. Three products quantified (3H24DMPA, 3H4MPA at 18.2 mg/L, 3H4MHA at 6.4 mg/L), n=3 for titers, up to seven media variants that are not fully crossed, single 48 h harvest. The genotype classes here exceed the usual four and need extra encoding: alongside deletions and plasmid PKS expression there are mini-Tn7 and BxB1 chromosomal integrations, C-terminal ssrA degron tags on FabD (LAA active, DAS intermediate, LDD inert), and acyltransferase-domain swaps in the LipPKS1 loading module. Accompanying RB-TnSeq is said to be at fit.genomics.lbl.gov, unconfirmed. A sibling deposit from the same group, PXD069956 (Engineering an Extremely Hybrid PKS for Adipic Acid Production, Klass et al., ACS Synth Biol, DOI 10.1021/acssynbio.5c00972, PMID 42339614), covers the same PKS platform in <i>P. putida</i> and <i>E. coli</i> if the modality is worth widening.
64	Alper 2005 lycopene Combinatorial design; lycopene content per cell dry weight	<i>E. coli</i>	4	64 designs	1 condition	192‡	192	K-12-KO	Two independent target-discovery routes, a genome-scale-model systematic search and a transposon-insertion combinatorial search, produced two disjoint gene sets whose exhaustive combination gave a defined set of 64 knockout strains, the best 8.5-fold over recombinant K-12 wild type and 2-fold over the engineered parent. The deletion set is in a K-12 background that also carries the heterologous lycopene pathway; transposon insertions were used for discovery, not as the final genotypes. Replicate counts were not verified.
65	ActiveOpt valine (Kumar 2021) Combinatorial design; valine elemental carbon yield (% of maximum theoretical yield from glucose plus acetate)	<i>E. coli</i>	4	93 RBS variants	1 condition	186‡	186	engineered-chassis+RBS	New experimental dataset reported in this paper: 39 plasmids expressing ilvBN*DE and ilvIH*C/C*-ygaZH with varied RBS strengths, tested as 89 pairwise plasmid combinations in strain PYR003a (BW25113 delta-aceE delta-gdhA delta-poxB delta-ldhA delta-recA), plus 4 new combinations chosen by ActiveOpt, so 93 genotype-linked yields; best strain 45% elemental carbon yield, 54.7% of maximum theoretical. A minimum of two biological replicate colonies per experiment, 48 h shake flask endpoint, no time course.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
66	Banerjee 2024 Production campaign; isoprenol titer plus growth and glucose consumption, shake flask and fed-batch	<i>P. putida</i>	4	19 deletions	3 conditions	171‡	171	KT2440-KO	The recalled 3.5 g/L fed-batch figure is confirmed in the abstract, against 1.1 g/L in batch and over tenfold above the starting strain. Eight deletion targets came from bilevel optimization and constrained minimal cut sets on iJN1462, and the best strain carries five deletions. The strain count is an estimate read from the preprint because the published version is paywalled. Proteomics here is targeted MRM of pathway enzymes only, not a discovery panel, so dimensionality stays 1. Linked assets: PXD039868 on PanoramaPublic plus six Experimental Data Depot studies at edd.jbei.org, which were not opened.
67	Wang 2018 phosphatase CRISPRi CRISPR library screen; lycopene content (and beta-carotene in follow-up strains)	<i>E. coli</i>	4	56 guides	1 condition	168‡	168	engineered-chassis	A CRISPRi knockdown library over 56 phosphatase-encoding genes built in a lycopene overproducer; 28 of the 56 knockdowns impaired lycopene synthesis, and combinatorial knockdowns plus knockouts were then tested in lycopene and beta-carotene overproducers. The gene set is small and fully named, so per-design values should be recoverable from the figures, but replicate counts were not verified so instances is an estimate.
68	Jain 2015 1,2-propanediol Production campaign; 1,2-propanediol titer (g/L) and yield (g/g glucose)	<i>E. coli</i>	4	20 designs	1 condition	60‡	120	engineered-chassis	I could not confirm the existence of a distinct growth-coupled 1,2-propanediol design-validation campaign: repeated searches surfaced only computational growth-coupling frameworks (OptKnock, minimal cut sets, gcFront) plus separate 1,2-PDO engineering papers, so the 1,2-propanediol row is filled by the best-documented reconstructable <i>E. coli</i> campaign, which reached 5.13 g/L and 0.48 g/g glucose via an optimal minimal enzyme set, flux channeling, raised NADH availability and improved anaerobic growth. The citation is sourced; the genotype and instance counts are unverified estimates, hence confidence recall.
69	Farasat 2014 neurosporene (ActiveOpt case 2) Combinatorial design; specific neurosporene productivity (microgram per gCDW per hour)	<i>E. coli</i>	4	101 RBS variants	1 condition	101	101	engineered-chassis+RBS	Kept as a row separate from the valine campaign because the design space is independent (crtEBI RBS library, different host and readout). The neurosporene measurements are Farasat et al. 2014's, not new work: 73 designed crtEBI expression constructs measured as exploration experiments plus 28 kinetic-model-designed extrapolation constructs, productivity rising from 196.3 to 286 microgram per gCDW per hour. ActiveOpt (Kumar 2021, doi 10.1016/j.ymben.2021.06.009) is a retrospective reanalysis of that dataset and generated no new neurosporene strains, so the primary citation is Farasat 2014.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
70	Lim 2020 ionic liquid ALE^B Tolerance / robustness; growth performance at high protic ionic liquid concentrations	<i>P. putida</i>	4	10 resequenced clones	3 conditions	90‡	90	evolved-WGS	Bibliography confirmed through CrossRef (Green Chemistry 22(17):5677-5690); the journal is not indexed in PubMed, so the PMID is deliberately left empty rather than guessed. Evolved strains grow at up to 4 percent triethanolammonium acetate and 8 percent triethylammonium hydrogen sulfate where the wild type cannot, with mutations in <i>relA</i> , <i>gacS</i> , <i>oprB</i> , <i>fleQ</i> , <i>tktA</i> and <i>uvrY</i> for minimal media and PP_5350, <i>emrE</i> , <i>oprD</i> and PP_5324 for the ionic-liquid-specific arms; PP_5350 (an RpiR-family regulator upregulating the glyoxylate cycle) and the <i>emrE</i> efflux pump were validated by reverse engineering. Marked BLOCKED because no sequencing deposit could be located: the publisher page returns HTTP 403, OSTI dropped the connection, and no BioProject or SRA accession surfaced anywhere, so the evolved genotypes are not retrievable even though the mutations are named in the text. Strain and instance counts are estimates.
71	Ling 2022 muconate Production campaign; cis,cis-muconate titer, molar yield and volumetric rate on glucose and xylose	<i>P. putida</i>	4	11 deletions	4 conditions	66‡	66	engineered-chassis	The best documented muconate campaign with per-strain titers: 11 strains carry titer or growth measurements and Table 1 lists 24 strains, across four carbon conditions in flask, plate reader and 0.5 L fed-batch, reaching 33.7 g/L at 0.18 g/L/h and 46 percent molar yield. PRJNA783062 confirmed by fetching the BioProject page, titled Sequencing of 5 ALE isolates on xylose (NREL, 2021-11-23), so only the genome resequencing is deposited and there is no proteomics accession. Genotype encoding is unusually mixed and does not fit a single modality: rational deletions and plasmid pathways sit alongside ALE-derived promoter mutations in the PP_2569 promoter, <i>xylE</i> point mutations (A62V, A455V) and a roughly 227.8 kb genome duplication from PP_5050 to PP_5242, none of which are designed edits.
72	Werner 2023 beta-ketoadipate Production campaign; beta-ketoadipate titer, productivity and molar yield from lignin-derived aromatics	<i>P. putida</i>	4	8 deletions	4 conditions	64‡	64	KT2440-KO	Eight strains with titers across four aromatic feeds (p-coumarate, ferulate, a 3:1 molar mixture, and corn stover alkaline pretreated liquor extractives), reaching 44.5 g/L at 1.15 g/L/h from model compounds and 25 g/L at 0.66 g/L/h from real corn stover aromatics; genotypes are deletions of <i>pcaIJ</i> , <i>crc</i> , <i>pobAR</i> , <i>lvaE</i> , <i>gacA</i> and <i>gacS</i> plus chromosomal overexpression at the <i>fpvA</i> locus. It has the best per-strain numerical table of this cluster, with every main-text data point tabulated in data S1. But there is no repository accession at all and strains are available only from NREL under a material transfer agreement, so <i>accession_confirmed</i> is false because no accession exists. One arm is informative as a negative: the <i>gacA</i> variants underperformed.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
73	Banerjee 2020 indigoidine Production campaign; indigoidine titer, rate and yield plus growth, scaled from deep-well plate to 2 L bioreactor	<i>P. putida</i>	4	2 guides	8 conditions	48†	48	KT2440+ guide	This is the growth-coupled indigoidine CRISPRi campaign, and it needs a CORRECTION to the recalled framing. The counts are right but the object is not: 63 constrained-minimal-cut-set solutions were analyzed, one feasible cut set hit 14 metabolic reactions, which mapped through GPRs to 16 single-copy genes, of which 2 (mqo-I and cynT) were dropped as essential by genome-wide RB-TnSeq, leaving 14 genes knocked down. Those 14 guides sit in ONE multiplexed dCpf1 gRNA array in a SINGLE strain. There are not 14 CRISPRi strains, so the genotype axis here is 2 (engineered versus empty-vector control), not 14, and anything that ingests this as a 14-strain panel is wrong. Conditions are 2 carbon sources crossed with 4 culture formats. RNA-seq is genome-wide (about 5,500 genes, estimated from the annotation, not stated); the proteomics is targeted SRM, so it is a selected reaction list rather than a discovery panel. BioProjects PRJNA580539 through PRJNA580574 (both endpoints resolve; the paper typos the upper bound); proteomics at PanoramaWeb, not opened.
74	Wang 2023 BATCH biosensor CRISPRi^B CRISPR library screen; biosensor fluorescence sort, then titer (mg/L) on selected sgRNA variants	<i>E. coli</i>	4	20 guides	2 conditions	40†	40	engineered- chassis	Doubly mismatched sgRNA pools give graded knockdown of 20 central-carbon-metabolism genes, screened with a PadR-based p-coumaric acid biosensor and an HpdR-based butyrate biosensor: pfkA plus ptsI variants raised p-coumarate 40.6% to 1308.6 mg/L from glycerol, and sucA or ldhA variants raised butyrate 19.0% and 25.2%. The total number of mismatched sgRNA variants in the pools was not confirmed, and the pooled sort yields no per-variant phenotype except for the few reconstructed winners, hence blocked. Conditions 2 counts the two product-biosensor campaigns.
75	Czajka 2022 Production campaign; indigoidine titer by A612 in DMSO plus growth rate	<i>P. putida</i>	4	6 guides	1 condition	36†	36	KT2440+ guide	The direct follow-up to Banerjee 2020 on the same 14-target multiplexed CRISPRi strain, adding an optimized Cpf1 ribosome binding site for a 1.6-fold gain, PHA-operon deletion strains, RNA-seq, targeted proteomics and 13C metabolic flux analysis. Six strains carry titers in one condition (M9, 10 g/L glucose, arabinose plus IPTG), n at least 6 for production profiling, sampled at 24, 48, 72 and 120 h. Kept as its own row rather than folded into Banerjee 2020 because the genotypes and the readout differ. Data availability names only public-registry.jbei.org with no folder number and states explicitly that everything else is in the manuscript and supplementary files, so there is no GEO, SRA, PRIDE or MassIVE deposit and nothing is machine-retrievable.

Table 5, continued

#	Dataset (class; phenotype)	Host	Tier	Genotypes	Env	Inst.	Meas.	Sequence basis	Why
76	Fong 2005 growth-coupled lactate Production campaign; lactate titer (g/L), growth rate, lactate secretion rate	<i>E. coli</i>	4	3 deletions	1 condition	11	33	K-12-KO	Three OptKnock-designed knockout strains were built and adaptively evolved to yield 11 evolved lactate-producing strains, growing on 2 g/L glucose at 37 C with titers 0.87 to 1.75 g/L and secretion rates coupled to growth rate; computational post-evolution growth-rate predictions were within 10% in 38 of 50 cases. Only 3 distinct engineered genotypes exist, so this is a small validation campaign rather than a library.
77	Yu 2016 PHBA Production campaign; para-hydroxybenzoic acid titer from glucose via chorismate, plus carbon yield	<i>P. putida</i>	4	7 deletions	1 condition	21†	21	KT2440-KO	The recalled six strains plus wild type is exact: S0 wild-type control and S1 through S6, built from markerless deletions of pobA, pheA, trpE and hexR combined with plasmid-borne <i>E. coli</i> <i>ubiC</i> and feedback-resistant <i>aroG</i> -D146N; S6 reached 1.73 g/L at 18.1 percent C-mol per C-mol. One defined medium, one 44 h endpoint, biological triplicates, which makes it the cheapest row here to ingest. A single S6 fed-batch run adds a short time series. Up to eight co-measured scalars exist per instance (PHBA, biomass, glucose, acetate, alpha-ketogluconate, pyruvate, lactate, succinate) if each counts as a phenotype. No proteomics, no transcriptomics, no repository deposit of any kind, so <code>accession_confirmed</code> is false because no accession exists to confirm.

Table 6: Sources for Table 5, in the same order. *Data* is where the per-record values live; an entry marked unconfirmed was not fetched live and must be checked before a loader is written. Every link is clickable.

#	Dataset, citation and link	Modality and readout	Data
1	Fuhrer 2017 Fuhrer T, Zampieri M, Sevin DC, Sauer U, Zamboni N. Genomewide landscape of gene-metabolome associations in <i>Escherichia coli</i> . Molecular Systems Biology 2017 doi:10.15252/msb.20167150	gene deletion endpoint or steady state	MassIVE MSV000078963; EBI BioStudies S-BSST5; Datasets EV1-EV4
2	Wetmore 2015 Wetmore KM, Price MN, Waters RJ, Lamson JS, He J, Hoover CA, Blow MJ, Bristow J, Butland G, Arkin AP, Deutschbauer A. Rapid quantification of mutant fitness in diverse bacteria by sequencing randomly bar-coded transposons. mBio 2015 doi:10.1128/mBio.00306-15	transposon insertion endpoint or steady state	http://genomics.lbl.gov/supplemental/rbarseq/ ; code at https://bitbucket.org/berkeleylab/feba (unconfirmed)
3	Mutalik 2020 Mutalik VK, Adler BA, Rishi HS, Piya D, Zhong C, Koskella B, Kutter EM, Calendar R, Novichkov PS, Price MN, Deutschbauer AM, Arkin AP. High-throughput mapping of the phage resistance landscape in <i>E. coli</i> . PLoS Biology 2020 doi:10.1371/journal.pbio.3000877	transposon insertion endpoint or steady state	SRA BioProject PRJNA645443; RB-TnSeq data at figshare 10.6084/m9.figshare.11413128; CRISPRi data at 10.6084/m9.figshare.11859216.v1; Dub-seq data at 10.6084/m9.figshare.11838879.v2 (unconfirmed)
4	Tong 2020 Tong M, French S, El Zahed SS, Ong WK, Karp PD, Brown ED. Gene dispensability in <i>Escherichia coli</i> grown in thirty different carbon environments. mBio 2020 doi:10.1128/mBio.02259-20	gene deletion sampled series	Supplementary Tables S1 to S6 via PMC7527729; CarPE web application at https://edbrownlab.shinyapps.io/CarPE/
5	PRECISE-1K Lamoureux CR, Decker KT, Sastry AV, Rychel K, Gao Y, McConn JL, Zielinski DC, Palsson BO. A multi-scale expression and regulation knowledge base for <i>Escherichia coli</i> . Nucleic Acids Research 2023 doi:10.1093/nar/gkad750	mixed endpoint or steady state	https://imodulondb.org (dataset <i>E.coli</i> PRECISE-1K); https://doi.org/10.5281/zenodo.8284223 ; https://github.com/SBRG/precise1k
6	Carruthers 2025 Carruthers DN, Kinnunen PC, Li Y, Chen Y, Gin JW, Yunus IS, et al. Automation and machine learning drive rapid optimization of isoprenol production in <i>Pseudomonas putida</i> . Nature Communications 2025 doi:10.1038/s41467-025-66304-8	CRISPRi endpoint or steady state	https://doi.org/10.5061/dryad.gtht76hzh
7	Lim 2022 putidaPRECISE321 Lim HG, Rychel K, Sastry AV, Bentley GJ, Mueller J, Schindel HS, et al. Machine-learning from <i>Pseudomonas putida</i> KT2440 transcriptomes reveals its transcriptional regulatory network. Metabolic Engineering 2022 doi:10.1016/j.ymben.2022.04.004	environment-only endpoint or steady state	https://github.com/SBRG/modulome_ppu

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
8	Borchert 2024 fModules Borchert AJ, Bleem AC, Lim HG, Rychel K, Dooley KD, Kellermeyer ZA, Hodges TL, Palsson BO, Beckham GT. Machine learning analysis of RB-TnSeq fitness data predicts functional gene modules in <i>Pseudomonas putida</i> KT2440. <i>mSystems</i> 2024 doi:10.1128/msystems.00942-23	transposon insertion endpoint or steady state	https://github.com/beckham-lab/fModule
9	Caglar 2017 Caglar MU, Houser JR, Barnhart CS, Boutz DR, Carroll SM, Dasgupta A, Lenoir WF, Smith BL, Sridhara V, Sydykova DK, Vander Wood D, Marx CJ, Marcotte EM, Barrick JE, Wilke CO. The <i>E. coli</i> molecular phenotype under different growth conditions. <i>Scientific Reports</i> 2017 doi:10.1038/srep45303	environment-only sampled series	GSE67402 and GSE94117 (GEO); PXD002140 and PXD005721 (PRIDE); flux data doi:10.18738/T8/UG3TUR
10	Nichols 2011 Nichols RJ, Sen S, Choo YJ, Beltrao P, Zietek M, Chaba R, Lee S, Kazmierczak KM, Lee KJ, Wong A, Shales M, Lovett S, Winkler ME, Krogan NJ, Typas A, Gross CA. Phenotypic landscape of a bacterial cell. <i>Cell</i> 2011 doi:10.1016/j.cell.2010.11.052	gene deletion endpoint or steady state	https://ecoliwiki.org/tools/chemgen/(coli_FinalData2.txt, 12.3 Mb; nichols.tgz, 61.5 Mb; conditions.txt); Cell Table S2 via PMC3060659
11	Goodall 2018 Goodall ECA, Robinson A, Johnston IG, Jabbari S, Turner KA, Cunningham AF, Lund PA, Cole JA, Henderson IR. The essential genome of <i>Escherichia coli</i> K-12. <i>mBio</i> 2018 doi:10.1128/mBio.02096-17	transposon insertion endpoint or steady state	ENA PRJEB24436 (unconfirmed)
12	Price 2018 Price MN, Wetmore KM, Waters RJ, Callaghan M, Ray J, Liu H, Kuehl JV, Melnyk RA, Lamson JS, Suh Y, Carlson HK, Esquivel Z, Sadeeshkumar H, Chakraborty R, Zane GM, Rubin BE, Wall JD, Visel A, Bristow J, Blow MJ, Arkin AP, Deutschbauer AM. Mutant phenotypes for thousands of bacterial genes of unknown function. <i>Nature</i> 2018 doi:10.1038/s41586-018-0124-0	transposon insertion endpoint or steady state	https://genomics.lbl.gov/supplemental/bigfit/html/Keio/(fit_logratios_good.tab, fit_t.tab, strain_fit.tab, all.poolcount); figshare 10.6084/m9.figshare.5134840 and 10.6084/m9.figshare.5134837; Fitness Browser https://fit.genomics.lbl.gov
13	Choe 2025 Choe D, Lee E, Kim K, Hwang S, Jeong KJ, Palsson BO, Cho BK, Cho S. Rapid identification of key antibiotic resistance genes in <i>E. coli</i> using high-resolution genome-scale CRISPRi screening. <i>iScience</i> 2025 doi:10.1016/j.isci.2025.112435	CRISPRi endpoint or steady state	ENA PRJEB33267 (<i>E. coli</i> genome-wide CRISPRi under multiple chemical treatments)
14	Yunus 2026 Yunus IS, Carruthers DN, Chen Y, Gin JW, Baidoo EEK, Petzold CJ, Garcia Martin H, Adams PD, Mukhopadhyay A, Lee TS. Predictive CRISPR-mediated gene downregulation for enhanced production of sustainable aviation fuel precursor in <i>Pseudomonas putida</i> . <i>Metabolic Engineering</i> 2026 doi:10.1016/j.ymben.2025.11.007	CRISPRi endpoint or steady state	(unconfirmed)

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
15	de Siqueira 2025 de Siqueira GMV, Srinivasan A, Chen Y, Gin JW, Petzold CJ, Lee TS, Guazzaroni ME, Eng T, Mukhopadhyay A. Alternate routes to acetate tolerance lead to varied isoprenol production from mixed carbon sources in <i>Pseudomonas putida</i> . <i>Applied and Environmental Microbiology</i> 2025 doi:10.1128/aem.02123-24	laboratory evolution sampled series	https://www.ebi.ac.uk/pride/archive/projects/PXD055153
16	Wang 2015 isoprenol tolerance Wang C, Yang L, Shah AA, Choi ES, Kim SW. Dynamic interplay of multidrug transporters with TolC for isoprenol tolerance in <i>Escherichia coli</i> . <i>Scientific Reports</i> 2015;5:16505 doi:10.1038/srep16505	gene deletion endpoint or steady state	Springer ESM 41598_2015_BFsrep16505_MOESM1_ESM.pdf (Tables S2, S3, S4); strains are Keio JW numbers, NBRP National Institute of Genetics
17	Lim 2025 isoprenol TALE Lim HG, Srinivasan A, Menchavez R, Yunus IS, Noh MH, White M, et al. Evolution-guided tolerance engineering of <i>Pseudomonas putida</i> KT2440 for production of the aviation fuel precursor isoprenol. <i>Metabolic Engineering</i> 2025 doi:10.1016/j.ymben.2025.05.007	laboratory evolution sampled series	https://www.ebi.ac.uk/pride/archive/projects/PXD054609
18	Tian 2019 isopentenol CRISPRi Tian T, Kang JW, Kang A, Lee TS. Redirecting metabolic flux via combinatorial multiplex CRISPRi-mediated repression for isopentenol production in <i>Escherichia coli</i> . <i>ACS Synthetic Biology</i> 2019;8:391-402 doi:10.1021/acssynbio.8b00429	multiplex CRISPRi knockdown sampled at 24 and 48 h, with the methods also stating 76 h against 72 h in the results	JBEI public registry (https://public-registry.jbei.org/), JPUB numbers per plasmid in Table 1; no sequence-data or value accession (unconfirmed)
19	Kang 2026 isoprenyl acetate Kang CW, Carruthers DN, McCauley J, Chen Y, Gin JW, Petzold CJ, Simmons BA, Lee TS. Multi-layered metabolic remodeling of <i>Pseudomonas putida</i> for efficient conversion of lignocellulosic sugars to the precursors of advanced aviation fuel. <i>Metabolic Engineering Communications</i> 2026 doi:10.1016/j.mec.2026.e00274	pathway plasmid sampled series	https://www.ebi.ac.uk/pride/archive/projects/PXD067010
20	Menasalvas 2025 Menasalvas J, Kulakowski S, Chen Y, Gin JW, Akyuz Turumtay E, Baral NR, et al. Biosensor-driven strain engineering reveals key cellular processes for maximizing isoprenol production in <i>Pseudomonas putida</i> . <i>Science Advances</i> 2025 doi:10.1126/sciadv.ady2677	gene deletion sampled series	https://doi.org/10.5061/dryad.sbcc2frjq
21	Rapp 2026 CRISPRi metabolome Rapp J, Verhulsdonk A, Garcke A, Stadelmann A, Farke N, Trossmann F, Kronenberger T, Alvarado A, Petras D, Link H. The metabolome of an <i>E. coli</i> CRISPRi library identifies benefits of minimal metabolite levels and targets for engineering. <i>Cell Systems</i> 2026 doi:10.1016/j.cels.2025.101518	CRISPRi endpoint or steady state	MassIVE MSV000095534 ('A metabolism-wide CRISPRi library expands the measurable <i>E.coli</i> metabolome -FI-MS dataset', submitted by H.Link, 2024-08-07); Zenodo https://zenodo.org/records/15640293

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
22	Schmidt 2022 nitrogen Schmidt M, Pearson AN, Incha MR, Thompson MG, Baidoo EEK, Kakumanu R, et al. Nitrogen Metabolism in <i>Pseudomonas putida</i> : Functional Analysis Using Random Barcode Transposon Sequencing. Applied and Environmental Microbiology 2022 doi:10.1128/aem.02430-21	transposon insertion endpoint or steady state	https://fit.genomics.lbl.gov/cgi-bin/org.cgi?orgId=Putida (unconfirmed)
23	Wang 2018 Wang T, Guan C, Guo J, Liu B, Wu Y, Xie Z, Zhang C, Xing XH. Pooled CRISPR interference screening enables genome-scale functional genomics study in bacteria with superior performance. Nature Communications 2018 doi:10.1038/s41467-018-04899-x	CRISPRi endpoint or steady state	Supplementary Data 05 to 10 of https://www.nature.com/articles/s41467-018-04899-x ; code at https://github.com/zhangchonglab/CRISPRi-functional-genomics-in-prokaryotes ; library at Addgene (Chong Zhang <i>E.coli</i> Genome-wide Inhibition Library) (unconfirmed)
24	Rousset 2018 Rousset F, Cui L, Siouvé E, Becavin C, Depardieu F, Bikard D. Genome-wide CRISPR-dCas9 screens in <i>E. coli</i> identify essential genes and phage host factors. PLoS Genetics 2018 doi:10.1371/journal.pgen.1007749	CRISPRi endpoint or steady state	ENA PRJEB28256 (secondary study ERP110440); supplementary tables S1 to S10
25	Shiver 2016 Shiver AL, Osadnik H, Kritikos G, Li B, Krogan N, Typas A, Gross CA. A chemical-genomic screen of neglected antibiotics reveals illicit transport of kasugamycin and blasticidin S. PLoS Genetics 2016 doi:10.1371/journal.pgen.1006124	gene deletion endpoint or steady state	Dryad 10.5061/dryad.f3kc0, mirrored at https://zenodo.org/records/5085832 (imagefile.tar 10.5 GB, irisfile.tar 65.1 MB, ImageKeyFinal.xls, InputFiles.tar); scripts at https://github.com/AnthonyShiverMicrobes/fitness_score
26	Thompson 2020 fatty acid and alcohol Thompson MG, Incha MR, Pearson AN, Schmidt M, Sharpless WA, Eiben CB, et al. Fatty Acid and Alcohol Metabolism in <i>Pseudomonas putida</i> : Functional Analysis Using Random Barcode Transposon Sequencing. Applied and Environmental Microbiology 2020 doi:10.1128/aem.01665-20	transposon insertion endpoint or steady state	https://fit.genomics.lbl.gov/cgi-bin/org.cgi?orgId=Putida (unconfirmed)
27	Borchert 2023 lignin tolerance Borchert AJ, Bleem A, Beckham GT. RB-TnSeq identifies genetic targets for improved tolerance of <i>Pseudomonas putida</i> towards compounds relevant to lignin conversion. Metabolic Engineering 2023 doi:10.1016/j.ymben.2023.04.007	transposon insertion endpoint or steady state	https://www.ncbi.nlm.nih.gov/bioproject/PRJNA856070
28	Schastnaya 2021 Schastnaya E, Raguz Nakic Z, Gruber CH, Doubleday PF, Krishnan A, Johns NI, Park J, Wang HH, Sauer U. Extensive regulation of enzyme activity by phosphorylation in <i>Escherichia coli</i> . Nature Communications 2021 doi:10.1038/s41467-021-25988-4	mixed endpoint or steady state	PXD027243 (PRIDE); MSV000087795 (MassIVE metabolomics)

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
29	Cui 2018 Cui L, Vigouroux A, Rousset F, Varet H, Khanna V, Bikard D. A CRISPRi screen in <i>E. coli</i> reveals sequence-specific toxicity of dCas9. <i>Nature Communications</i> 2018 doi:10.1038/s41467-018-04209-5	CRISPRi endpoint or steady state	Supplementary Data 4 of https://www.nature.com/articles/s41467-018-04209-5 (no repository accession) (unconfirmed)
30	Schmidt 2016 Schmidt A, Kochanowski K, Vedelaar S, Ahrne E, Volkmer B, Callipo L, Knoops K, Bauer M, Aebersold R, Heinemann M. The quantitative and condition-dependent <i>Escherichia coli</i> proteome. <i>Nature Biotechnology</i> 2016 doi:10.1038/nbt.3418	environment-only endpoint or steady state	PXD000498 (ProteomeXchange/PRIDE, DOI 10.6019/PXD000498)
31	Butland 2008 Butland G, Babu M, Diaz-Mejia JJ, Bohdana F, Phanse S, Gold B, Yang W, Li J, Gagarinova AG, Pogoutse O, Mori H, Wanner BL, Lo H, Wasniewski J, Christopoulos C, Ali M, Venn P, Safavi-Naini A, Sourour N, Caron S, Choi JY, Laigle L, Nazarians-Armavil A, Deshpande A, Joe S, Datsenko KA, Yamamoto N, Andrews BJ, Boone C, Ding H, Sheikh B, Moreno-Hagelsieb G, Greenblatt JF, Emili A. eSGA: <i>E. coli</i> synthetic genetic array analysis. <i>Nature Methods</i> 2008 doi:10.1038/nmeth.1239	double deletion endpoint or steady state	(unconfirmed)
32	Mori 2021 Mori M, Zhang Z, Banaei-Esfahani A, Lalanne JB, Okano H, Collins BC, Schmidt A, Schubert OT, Lee DS, Li GW, Aebersold R, Hwa T, Ludwig C. From coarse to fine: the absolute <i>Escherichia coli</i> proteome under diverse growth conditions. <i>Molecular Systems Biology</i> 2021 doi:10.15252/msb.20209536	environment-only endpoint or steady state	PASS01421 (SWATHAtlas spectral library); Datasets EV1-EV12
33	Ishii 2007 Ishii N, Nakahigashi K, Baba T, Robert M, Soga T, Kanai A, Hirasawa T, Naba M, Hirai K, Hoque A, Ho PY, Kakazu Y, Sugawara K, Igarashi S, Harada S, Masuda T, Sugiyama N, Togashi T, Hasegawa M, Takai Y, Yugi K, Arakawa K, Iwata N, Toya Y, Nakayama Y, Nishioka T, Shimizu K, Mori H, Tomita M. Multiple high-throughput analyses monitor the response of <i>E. coli</i> to perturbations. <i>Science</i> 2007 doi:10.1126/science.1132067	mixed endpoint or steady state	Science supporting online material; no repository accession found (unconfirmed)
34	Campos 2018 Campos M, Govers SK, Irnov I, Dobihal GS, Cornet F, Jacobs-Wagner C. Genomewide phenotypic analysis of growth, cell morphogenesis, and cell cycle events in <i>Escherichia coli</i> . <i>Molecular Systems Biology</i> 2018 doi:10.15252/msb.20177573	gene deletion endpoint or steady state	Dataset EV2 (corrected and normalized scores) with the article at https://www.embopress.org/doi/full/10.15252/msb.20177573

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
35	Typas 2008 Typas A, Nichols RJ, Siegele DA, Shales M, Collins SR, Lim B, Braberg H, Yamamoto N, Takeuchi R, Wanner BL, Mori H, Weissman JS, Krogan NJ, Gross CA. High-throughput, quantitative analyses of genetic interactions in <i>E. coli</i> . <i>Nature Methods</i> 2008 doi:10.1038/nmeth.1240	double deletion endpoint or steady state	Supplementary Tables 2A and 2B via PMC2700713; author manuscript at https://escholarship.org/content/qt0b4526ns/qt0b4526ns.pdf (unconfirmed)
36	Girgis 2009 Girgis HS, Hottes AK, Tavazoie S. Genetic architecture of intrinsic antibiotic susceptibility. <i>PLoS ONE</i> 2009 doi:10.1371/journal.pone.0005629	transposon insertion endpoint or steady state	PLoS ONE Supporting Information Datasets S1 to S5 at https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0005629 (no repository accession) (unconfirmed)
37	Banerjee 2025 Banerjee D, Menasalvas J, Chen Y, Gin JW, Baidoo EEK, Petzold CJ, Eng T, Mukhopadhyay A. Addressing genome scale design tradeoffs in <i>Pseudomonas putida</i> for bioconversion of an aromatic carbon source. <i>npj Systems Biology and Applications</i> 2025 doi:10.1038/s41540-024-00480-z	promoter variant sampled series	https://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PXD050285
38	Fang 2025 FFA CRISPRi-FACS Fang L, Hao X, Fan J, Liu X, Chen Y, Wang L, Huang X, Song H, Cao Y. Genome-scale CRISPRi screen identifies pcnB repression conferring improved physiology for overproduction of free fatty acids in <i>Escherichia coli</i> . <i>Nature Communications</i> 2025 doi:10.1038/s41467-025-58368-3	CRISPRi endpoint or steady state	https://www.nature.com/articles/s41467-025-58368-3 (unconfirmed)
39	Babu 2014 Babu M, Arnold R, Bundalovic-Torma C, Gagarinova A, Wong KS, Kumar A, Stewart G, Samanfar B, Aoki H, Wagih O, Vlasblom J, Phanse S, Lad K, Yeou Hsiung Yu A, Graham C, Jin K, Brown E, Golshani A, Kim P, Moreno-Hagelsieb G, Greenblatt J, Houry WA, Parkinson J, Emili A. Quantitative genome-wide genetic interaction screens reveal global epistatic relationships of protein complexes in <i>Escherichia coli</i> . <i>PLoS Genetics</i> 2014 doi:10.1371/journal.pgen.1004120	double deletion endpoint or steady state	Table S2 (XLS, high-confidence epistatic gene pairs) at https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1004120
40	Gupta 2024 turnover Gupta M, Johnson ANT, Cruz ER, Costa EJ, Guest RL, Li SHJ, Hart EM, Nguyen T, Stadlmeier M, Bratton BP, Silhavy TJ, Wingreen NS, Gitai Z, Wuhr M. Global protein turnover quantification in <i>Escherichia coli</i> reveals cytoplasmic recycling under nitrogen limitation. <i>Nature Communications</i> 2024 doi:10.1038/s41467-024-49920-8	environment-only sampled series	PXD042444 (ProteomeXchange/PRIDE); code at https://github.com/wuhrlab/ProteinTurnoverEcoli
41	MCF2Chem 2023 Cai P, Liu S, Zhang D, Hu QN. MCF2Chem: A manually curated knowledge base of biosynthetic compound production. <i>Biotechnology for Biofuels and Bioproducts</i> 2023 doi:10.1186/s13068-023-02419-8	mixed endpoint or steady state	https://mcf.lifesynter.com (unconfirmed)

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
42	CeCaFDB flux compendium Zhang Z, Shen T, Rui B, et al. CeCaFDB: a curated database for the documentation, visualization and comparative analysis of central carbon metabolic flux distributions explored by ¹³ C-fluxomics. Nucleic Acids Research 2015 doi:10.1093/nar/gku1137	mixed endpoint or steady state	http://www.cecafdb.org (unconfirmed)
43	Thompson 2019 lysine Thompson MG, Blake-Hedges JM, Cruz-Morales P, Barajas JF, Curran SC, Eiben CB, et al. Massively Parallel Fitness Profiling Reveals Multiple Novel Enzymes in Pseudomonas putida Lysine Metabolism. mBio 2019 doi:10.1128/mbio.02577-18	transposon insertion endpoint or steady state	https://fit.genomics.lbl.gov/cgi-bin/org.cgi?orgId=Putida (unconfirmed)
44	Rand 2017 Putida_ML5 library Rand JM, Pisithkul T, Clark RL, Thiede JM, Mehrer CR, Agnew DE, et al. A metabolic pathway for catabolizing levulinic acid in bacteria. Nature Microbiology 2017 doi:10.1038/s41564-017-0028-z	transposon insertion endpoint or steady state	https://fit.genomics.lbl.gov/cgi-bin/org.cgi?orgId=Putida (unconfirmed)
45	Balakrishnan 2022 Balakrishnan R, Mori M, Segota I, Zhang Z, Aebersold R, Ludwig C, Hwa T. Principles of gene regulation quantitatively connect DNA to RNA and proteins in bacteria. Science 2022 doi:10.1126/science.abk2066	mixed endpoint or steady state	PXD014948 (PRIDE); GSE205717 (GEO RNA-seq); PASS01421 (SWATHAtlas)
46	Brunk 2016 Brunk E, George KW, Alonso-Gutierrez J, Thompson M, Baidoo E, Wang G, Petzold CJ, McCloskey D, Monk J, Yang L, O'Brien EJ, Batth TS, Martin HG, Feist A, Adams PD, Keasling JD, Palsson BO, Lee TS. Characterizing Strain Variation in Engineered E. coli Using a Multi-Omics-Based Workflow. Cell Systems 2016 doi:10.1016/j.cels.2016.04.004	pathway plasmid sampled series	https://escholarship.org/uc/item/2k69b9zr (accepted manuscript plus supplementary files; analysis as iPython notebooks) (unconfirmed)
47	Thompson 2019 valerolactam Thompson MG, Valencia LE, Blake-Hedges JM, Cruz-Morales P, Velasquez AE, Pearson AN, et al. Omics-driven identification and elimination of valerolactam catabolism in Pseudomonas putida KT2440 for increased product titer. Metabolic Engineering Communications 2019 doi:10.1016/j.mec.2019.e00098	transposon insertion endpoint or steady state	https://fit.genomics.lbl.gov/cgi-bin/org.cgi?orgId=Putida (unconfirmed)
48	D2Cell 2026 Li X, et al. Leveraging large language models for metabolic engineering design. Trends in Biotechnology (preprint: bioRxiv) 2026 doi:10.1101/2024.09.09.612023	mixed endpoint or steady state	https://zenodo.org/records/18240770 (Cell_factory_dataset_Qwen_110b.xlsx, 11.4 MB); code https://github.com/LiLabTsinghua/D2Cell ; web server https://digitallifethu.com/d2cell

Table 6, continued

#	Dataset, citation and link	Modality and readout	Data
49	Li 2014 Li GW, Burkhardt D, Gross C, Weissman JS. Quantifying absolute protein synthesis rates reveals principles underlying allocation of cellular resources. Cell 2014 doi:10.1016/j.cell.2014.02.033	environment-only endpoint or steady state	GSE53767 (GEO); browsable table at http://ecoliwiki.net/tools/proteome/
50	Potel 2018 phosphoproteome Potel CM, Lin MH, Heck AJR, Lemeer S. Widespread bacterial protein histidine phosphorylation revealed by mass spectrometry-based proteomics. Nature Methods 2018 doi:10.1038/nmeth.4580	environment-only endpoint or steady state	PXD008369 (ProteomeXchange, title 'Widespread protein histidine phosphorylation in bacteria')

5 What each row maps onto in the supported set ✗

A bacterial row is cheap or expensive depending on whether the substrate already holds its record type. Table 7 names, for every row, the built *S. cerevisiae* dataset it mirrors and why. A named analog means the loader writes a record the schema already validates, against a different reference genome, so the work is retrieval and provenance. A row with no analog is naming a phenotype the substrate has never held.

Table 7: What each bacterial row maps onto in the supported set. A named analog means the loader writes a record type the schema already holds, against a different reference genome, so the work is the retrieval and the provenance rather than a new phenotype class. *What the schema still needs* is per-row and excludes the two blockers every row shares (Table 4); a dash means those two are the whole cost.

#	Bacterial row	Built yeast analog	Why it is the analog	What the schema still needs
1	Fuhrer 2017 <i>E. coli</i>	Mulleder 2016 amino-acid metabolome	whole deletion collection metabolite profiling	–
2	Wetmore 2015 <i>E. coli</i>	Hillenmeyer 2008 HIP/HOP	pooled barcoded competitive fitness assay	–
3	Mutalik 2020 <i>E. coli</i>	Hillenmeyer 2008 HIP/HOP	pooled barcoded mutants across selective conditions	–
4	Tong 2020 <i>E. coli</i>	Smith 2006 chemogenomic	colony growth per deletion per condition	–
5	PRECISE-1K <i>E. coli</i>	Kemmeren 2014 deletion transcriptome	deletion-and-condition expression compendium	–
6	Carruthers 2025 <i>P. putida</i>	Lian 2019 CRISPR-AID	multiplexed guide combinations scored by product	–
7	Lim 2022 putidaPRECISE321 <i>P. putida</i>	Caudal 2024 pan-transcriptome	aggregated RNA-seq compendium across many backgrounds	–
8	Borchert 2024 fModules <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant fitness across hundreds of conditions	–
9	Caglar 2017 <i>E. coli</i>	Zelezniak 2018 proteome and metabolome	two layers on identical samples	–
10	Nichols 2011 <i>E. coli</i>	Hillenmeyer 2008 HIP/HOP	deletion collection across many chemicals	–
11	Goodall 2018 <i>E. coli</i>	SGD essentiality	genome-wide essential gene call set	–
12	Price 2018 <i>E. coli</i>	Hillenmeyer 2008 HIP/HOP	pooled barcoded fitness across many conditions	–
13	Choe 2025 <i>E. coli</i>	Smith 2016 CRISPRi	pooled CRISPRi knockdown across drugs	–
14	Yunus 2026 <i>P. putida</i>	Lian 2019 CRISPR-AID	predicted multiplexed knockdowns scored by titer	–
15	de Siqueira 2025 <i>P. putida</i>	Mormino 2022 CRISPRi acetic acid	acetate tolerance with a product readout	–
16	Wang 2015 isoprenol tolerance <i>E. coli</i>	Lopez 2024 isobutanol	deletion collection scored for an alcohol phenotype	–
17	Lim 2025 isoprenol TALE <i>P. putida</i>	Mormino 2022 CRISPRi acetic acid	inhibitor tolerance scored by growth	–
18	Tian 2019 isopentenol CRISPRi <i>E. coli</i>	Lian 2019 CRISPR-AID	multiplexed guide arrays over one gene set	a guide-array perturbation carrying more than one target at a graded induction level, and a titer measured on a chassis whose pathway is plasmid-borne rather than integrated
19	Kang 2026 isoprenyl acetate <i>P. putida</i>	Lopez 2024 isobutanol	engineered strain series producing an alcohol ester	–

Table 7, continued

#	Bacterial row	Built yeast analog	Why it is the analog	What the schema still needs
20	Menasalvas 2025 <i>P. putida</i>	Lopez 2024 isobutanol	iterative engineered panel, alcohol titer	–
21	Rapp 2026 CRISPRi metabolome <i>E. coli</i>	Zelezniak 2018 metabolome	both put metabolite concentrations and a second molecular layer on one strain axis	–
22	Schmidt 2022 nitrogen <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant fitness across nitrogen sources	–
23	Wang 2018 <i>E. coli</i>	Smith 2016 CRISPRi	pooled CRISPRi guide fitness screen	–
24	Rousset 2018 <i>E. coli</i>	Smith 2016 CRISPRi	pooled CRISPRi guide fitness screen	–
25	Shiver 2016 <i>E. coli</i>	Wildenhain 2015 drug tolerance	neglected-compound panel against deletion collection	–
26	Thompson 2020 fatty acid and alcohol <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant fitness across a carbon-source panel	–
27	Borchert 2023 lignin tolerance <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant fitness across chemical stresses	–
28	Schastnaya 2021 <i>E. coli</i>	Mulleder 2016 amino-acid metabolome	point-mutant panel, metabolite readout	–
29	Cui 2018 <i>E. coli</i>	Smith 2016 CRISPRi	pooled CRISPRi guide fitness screen	–
30	Schmidt 2016 <i>E. coli</i>	Messner 2023 proteome	both give panel-wide absolute proteome depth	–
31	Butland 2008 <i>E. coli</i>	Costanzo 2016 SGA	conjugation analog of yeast SGA	–
32	Mori 2021 <i>E. coli</i>	Messner 2023 proteome	absolute proteome across a large condition panel	–
33	Ishii 2007 <i>E. coli</i>	Zelezniak 2018 proteome and metabolome	several layers, one strain panel	–
34	Campos 2018 <i>E. coli</i>	Kemmeren 2014 deletion transcriptome	high-dimensional profile per deletion strain	–
35	Typas 2008 <i>E. coli</i>	Costanzo 2016 SGA	conjugation analog of yeast SGA	–
36	Girgis 2009 <i>E. coli</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant library across drug panel	–
37	Banerjee 2025 <i>P. putida</i>	Zelezniak 2018 proteome	engineered panel paired with quantitative proteomes	–
38	Fang 2025 FFA CRISPRi-FACS <i>E. coli</i>	Xue 2025 free fatty acids	both read free fatty acid output as the label on a perturbed strain	–
39	Babu 2014 <i>E. coli</i>	Costanzo 2016 SGA	conjugation digenic epistasis colony scores	–
40	Gupta 2024 turnover <i>E. coli</i>	Messner 2023 proteome	proteome-wide per-protein quantitative depth	–
41	MCF2Chem 2023 <i>E. coli</i>	Ozaydin 2013 beta-carotene screen	both score a small-molecule product per engineered strain over a genome-scale perturbation panel	–
42	CeCaFDB flux compendium <i>E. coli</i>	–	No built row measures this.	–

Table 7, continued

#	Bacterial row	Built yeast analog	Why it is the analog	What the schema still needs
43	Thompson 2019 lysine <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled deletion fitness across nutrient conditions	–
44	Rand 2017 Putida_ML5 library <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	the pooled barcoded mutant collection itself	–
45	Balakrishnan 2022 <i>E. coli</i>	Zelezniak 2018 proteome and metabolome	two layers, same samples, joinable	–
46	Brunk 2016 <i>E. coli</i>	Zelezniak 2018 metabolome	both put metabolite concentrations and a second molecular layer on one strain axis	–
47	Thompson 2019 valerolactam <i>P. putida</i>	Hillenmeyer 2008 HIP/HOP	pooled mutant fitness on defined carbon sources	–
48	D2Cell 2026 <i>E. coli</i>	Ozaydin 2013 beta-carotene screen	both score a small-molecule product per engineered strain over a genome-scale perturbation panel	–
49	Li 2014 <i>E. coli</i>	Messner 2023 proteome	genome-wide absolute protein-level quantification	–
50	Potel 2018 phosphoproteome <i>E. coli</i>	Messner 2023 proteome	deep single-organism protein-level mass spectrometry	–

6 Before a loader is written ✗

Every figure in Table 5 is an axis count rather than a verified record count, and the qualifications below decide which rows can be trusted at their stated size.

Instance counts marked † are products, not counts. A screen reported as n strains against k conditions enters as nk instances. Real screens drop strains per condition, so each such figure is an upper bound. The ones marked ‡ are weaker still, being order-of-magnitude estimates for libraries whose exact size was not confirmed. Neither should be quoted as a dataset size without opening the source.

Dimensionality is the other half of every rank, so an estimated depth moves the order. Because rows are ranked on instances times dimensionality, a wrong panel depth moves a row directly rather than at the margin. The generated table records whether each depth was read from a source or estimated, and the summary counts the split. A row whose position looks surprising should have its depth checked before its instance count.

A corpus is not a dataset. Rows marked **A** are curated literature collections. Their record counts are counts of reported observations extracted from other papers, so they are not net new until the corpus is resolved to primary sources, and several of those sources are already elsewhere in this table. They are ranked because their aggregate size is real and useful for discovery, and they are flagged because counting them as independent datasets would double-count.

Rows marked B have no released per-record values. A screen whose released form is a figure or a table of selected hits is not ingestible as a genotype-to-phenotype dataset, whatever its reputation. These stay in the table with the flag rather than being dropped, because an author request or a supplementary re-check can change the state.

6.1 The transposon rows overlap each other, and the largest is a superset ✗

Several of the highest-ranking rows draw on the same barcoded libraries, so ingesting them in rank order would load the same measurements more than once. This is the one place where the ranking and the build order should come apart.

For *P. putida*, one released fitness matrix is dense and machine-readable at 4,732 genes by 332 samples with no missing values, and it is a superset of the other transposon-fitness rows in this table. Those rows should be de-duplicated against it rather than ingested in parallel, which turns four loaders into one plus a provenance record of which source experiments the matrix covers. The library behind them is one library, not several: the reported insertion counts belong to a single barcoded collection of 185,401 insertions, 152,810 of them in genes covering 5,213 of 5,661 genes, and a smaller figure quoted elsewhere is fitness coverage rather than library size.

For *E. coli*, the same relation holds between the method paper that introduced the barcoded library and the later compendium built on it: the compendium is a superset of that paper's *E. coli* experiments. The deletion-collection rows have the cleaner relationship, since they share a strain axis without sharing measurements, and that shared axis is what makes them joinable rather than redundant.

6.2 Access findings that change what a loader can do ✗

Four findings are specific enough to name, because each one changes the work rather than the confidence.

One *P. putida* campaign is deposited as seven separate proteomics accessions, one per design cycle. Ingesting the one accession a reader finds first would drop six sevenths of the proteome. The campaign is one dataset with linked assets, and its loader has to enumerate all seven.

No *P. putida* knockdown library in this table releases a guide-by-sample matrix. The library itself is real, roughly three guides per gene over 5,591 coding sequences, but the deposit holds baseline per-guide counts, a lost-guide list, per-round enriched hit lists and representative rather than per-replicate sequencing. A per-guide fitness matrix cannot be reconstructed from it, so the row enters on its per-strain production measurements and not as a pooled screen.

Two accessions did not resolve when checked. One resequencing project identifier returns nothing at the sequence archives, so the mutation calls exist only in a supplementary table, and one deletion-collection portal has moved host. Neither is a reason to drop a row, and both are a reason the provenance record has to pin the file that was actually read.

One fitness portal is behind a challenge that blocks scripted retrieval. Four rows whose only data location is that portal are therefore unconfirmed at the accession level, and the coverage figures circulating for it come from archived snapshots rather than a live read. They are recorded as unconfirmed rather than quoted as current.

7 What was dropped x

Table 8: Considered and dropped. *no-sequence* is the hard gate: without a route to the strain’s genomic content there is no genotype to map a phenotype from. *no-per-record-data* is a real experiment whose released form is a figure or a summary rather than per-strain values.

Dataset or group	Rule	Reason
Multi-host cell-factory capacity resource (modeled yields)	not-a-dataset	Genome-scale model predictions of production capacity across host and product combinations. Every value is computed rather than measured, so it belongs in a model-derived partition and must not enter an experimental instance or dataset count. Useful for negative sampling and as a feature source, which is a different role from training data
Simulated FSEOF rows inside the curated engineering corpora	not-a-dataset	The larger metabolic-engineering corpora carry both experimentally reported modifications and rows generated by flux-scanning enforced objective flux. The simulated rows are predictions of what an intervention would do and are excluded or separately typed; mixing them into an experimental count is the specific error these corpora invite
Adaptive laboratory evolution without resequencing	no-sequence	An evolved population differs from its parent at unmapped positions, so there is no genotype to map a phenotype from. Only resequenced clones and explicitly reconstructed mutations from such a campaign are admitted, which is why the tolerance-evolution rows enter on their sequenced isolates rather than on their lineages
Genome shuffling, chemical and UV mutagenesis strain panels	no-sequence	Same gate as unsequenced evolution, and it bites harder: the selected isolate carries an unknown number of edits at unknown positions. No amount of phenotypic value substitutes for a missing genotype
Other bacterial hosts (B. subtilis, C. glutamicum, Z. mobilis, cyanobacteria)	off-host	Real and in several cases large, but outside the two hosts this list covers. The generalization axis is single-cell bioproduction hosts, so these become candidates once either bacterial host is actually served and the per-host schema and genome-tier cost has been paid once
EcoCyc and the curated annotation databases	not-a-dataset	EcoCyc releases per-class flat files covering 4,688 genes, 479 pathways and 6,124 regulatory interactions for the reference strain (BioCyc v30.0, checked live). Those are annotations of one genome rather than measured genotype-by-environment records, so under a ranking on instances times dimensionality the curation would outrank real screens. It belongs in the reference layer the graph already has, alongside the gene ontology, not in a candidate list of experiments
Review and perspective tables of reported titers	not-a-dataset	A titer quoted in a review is a second-hand number without the strain construction, the medium or the replicate structure that makes it a record. The primary paper is the candidate; the review is a discovery index, and the curated corpora already serve that role better

8 Summary statistics ✗

Table 9: Summary of the fifty, split at the tranche-1 line. *Tr. 1* and *Tr. 2* count rows; *Genotypes*, *Instances* and *Meas.* sum the row axes over all fifty and are lower bounds, since a row with no count contributes nothing. The fifty name 52 joins, 49 of them to a dataset already built.

	Tr. 1	Tr. 2	Genotypes	Instances	Meas.
<i>By host</i>					
E. coli	12	23	2,968,285	3.0×10^7	3.0×10^8
P. putida	8	7	214,741	2.4×10^6	6.4×10^6
<i>By tier</i>					
tier 1	8	9	2,666,704	3.1×10^7	3.0×10^8
tier 2	2	6	311,385	1.2×10^6	1.4×10^6
tier 3	3	7	204,653	4.2×10^4	7.1×10^6
tier 4	7	8	284	1.9×10^3	3.9×10^6
<i>By class</i>					
Aggregation / support	0	3	17,022	1.7×10^4	6.6×10^4
CRISPR library screen	1	5	303,848	1.2×10^6	1.7×10^6
Combinatorial design	1	0	24	84	168
Fitness / chemical genomics	2	2	15,977	1.6×10^6	9.8×10^6
Genetic interaction	0	3	918,141	2.9×10^5	2.9×10^5
Metabolome / flux	1	1	3,896	3.4×10^4	2.6×10^8
Modality / backbone	0	1	185,401	1.9×10^4	1.9×10^4
Multi-omics campaign	1	3	37	440	1.5×10^6
Production campaign	4	1	694	3.1×10^3	2.2×10^6
Proteome	0	3	7	134	3.1×10^5
Tolerance / robustness	3	0	100	271	271
Transcriptome	2	0	93	1.4×10^3	6.2×10^6
Translation / turnover	0	2	2	15	4.8×10^4
Transposon fitness	5	6	1,737,784	2.9×10^7	2.9×10^7
<i>By sequence basis</i>					
K-12+guide	1	5	303,848	1.2×10^6	1.7×10^6
K-12+transposon	4	1	1,709,208	2.7×10^7	2.7×10^7
K-12-KO	5	3	19,932	1.7×10^6	2.7×10^8
K-12-KO x KO	0	3	918,141	2.9×10^5	2.9×10^5
KT2440+guide	2	0	492	1.5×10^3	2.1×10^6
KT2440+promoter	0	1	12	36	7.2×10^4
KT2440+transposon	1	6	213,977	2.4×10^6	2.4×10^6
engineered-chassis	2	2	288	2.1×10^3	2.2×10^5
engineered-chassis+guide	1	0	24	84	168
evolved-WGS	2	0	53	177	177
reference-only	2	9	17,051	1.8×10^4	3.6×10^6
<i>Other attributes</i>					
carries a time axis	7	3	4,086	1.2×10^5	9.7×10^6
mirrors a built yeast dataset	20	29	3,183,026	3.2×10^7	3.1×10^8
needs a phenotype class the substrate lacks	0	1	0	297	2.3×10^4
has a join to a built dataset	20	29	3,183,026	3.2×10^7	3.1×10^8
figures sourced this pass	20	28	3,027,610	3.2×10^7	3.1×10^8
figures from recall, to confirm	0	2	155,416	1.6×10^5	1.6×10^5
data location confirmed live	15	15	813,316	4.8×10^6	2.8×10^8
per-record values not released	1	3	302,765	3.0×10^5	3.0×10^5
a corpus, not net new until split	2	3	21,771	1.6×10^6	3.4×10^6
instances a reported count	6	8	757,736	1.8×10^6	2.7×10^8
instances a product of the axes	12	16	1,584,415	3.0×10^7	3.3×10^7
instances an estimate	2	6	840,875	2.5×10^5	3.7×10^5
dimensionality reported, not estimated	16	24	2,958,417	3.2×10^7	3.0×10^8
Total	20	30	3,183,026	3.2×10^7	3.1×10^8

Table 10: Candidates by class and by host, split at the two tranche lines. *Genotypes*, *Instances* and *Meas.* sum the per-row axes over all tranches; a row with no count contributes nothing, so every total is a lower bound.

	Tr. 1	Tr. 2	Reserve	Genotypes	Instances	Meas.
Aggregation / support	0	3	1	18,222	1.9×10^4	7.0×10^4
CRISPR library screen	1	5	3	304,032	1.2×10^6	1.7×10^6
Combinatorial design	1	0	5	677	1.1×10^3	1.2×10^3
Fitness / chemical genomics	2	2	0	15,977	1.6×10^6	9.8×10^6
Genetic interaction	0	3	0	918,141	2.9×10^5	2.9×10^5
Metabolome / flux	1	1	1	3,987	3.5×10^4	2.6×10^8
Modality / backbone	0	1	1	189,386	2.3×10^4	2.3×10^4
Multi-omics campaign	1	3	3	1,117	1.7×10^3	1.5×10^6
Production campaign	4	1	10	830	3.8×10^3	2.2×10^6
Proteome	0	3	0	7	134	3.1×10^5
Tolerance / robustness	3	0	2	215	781	781
Transcriptome	2	0	0	93	1.4×10^3	6.2×10^6
Translation / turnover	0	2	0	2	15	4.8×10^4
Transposon fitness	5	6	1	1,741,530	2.9×10^7	2.9×10^7
<i>The same rows, by host</i>						
E. coli	12	23	17	2,979,247	3.0×10^7	3.0×10^8
P. putida	8	7	10	214,969	2.4×10^6	6.4×10^6
Total	20	30	27	3,194,216	3.2×10^7	3.1×10^8